

Statistical Inference for Linear Stochastic Approximation with Richardson-Romberg Extrapolation

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Abstract

Your abstract.

1. Introduction

Stochastic approximation (SA) algorithms are a cornerstone of modern computational statistics, optimization, and reinforcement learning. Introduced by Robbins and Monro (1951), these iterative procedures provide a principled way to find roots of equations or optimize objectives when only noisy observations are available. A particularly important subclass is the *linear stochastic approximation* (LSA) algorithm, which arises naturally in temporal-difference (TD) learning, policy evaluation, and stochastic gradient descent for linear models.

In this work, we study the LSA recursion with a *constant step size* $\alpha > 0$:

$$\theta_k^{(\alpha)} = \theta_{k-1}^{(\alpha)} - \alpha \{A(Z_k)\theta_{k-1}^{(\alpha)} - b(Z_k)\}, \quad k \geq 1,$$

where $\{Z_k\}_{k \in \mathbb{N}}$ is a time-homogeneous Markov chain on a measurable space $(Z, \mathcal{Z})$ with transition kernel Q and unique invariant distribution π . The mappings $A : Z \rightarrow \mathbb{R}^{d \times d}$ and $b : Z \rightarrow \mathbb{R}^d$ are measurable functions satisfying $\bar{A} := \int_Z A(z) d\pi(z)$ and $\bar{b} := \int_Z b(z) d\pi(z)$. Assuming that $-\bar{A}$ is a *Hurwitz matrix* (all eigenvalues have strictly negative real parts), the target parameter $\theta^* = \bar{A}^{-1}\bar{b}$ is uniquely defined.

The *Polyak–Ruppert averaging* procedure (Polyak, 1990; Ruppert, 1988) provides an effective variance reduction technique. Given a burn-in period $n_0 \geq 0$, the averaged iterate is defined as

$$\bar{\theta}_n^{(\alpha)} = \frac{1}{n - n_0} \sum_{k=n_0}^{n-1} \theta_k^{(\alpha)}.$$

1.1. Bias of constant step-size iterates

The use of a constant step size $\alpha > 0$ offers several practical advantages: it enables geometrically fast forgetting of the initial condition (Dieuleveut, Durmus, and Bach, 2020) and simplifies hyperparameter tuning compared to diminishing step-size schedules. However, unlike the classical regime $\alpha_k \rightarrow 0$ with $\sum \alpha_k = \infty$ and $\sum \alpha_k^2 < \infty$, a constant step size produces iterates that converge only *in distribution* to a stationary measure Π_α , rather than almost surely to θ^* . The stationary expectation $\mathbb{E}[\theta_\infty^{(\alpha)}]$ is generally *biased* with respect to θ^* , and this bias cannot be eliminated by Polyak–Ruppert averaging alone.

As shown in Levin, Naumov, and Samsonov (2025), the stationary bias has a leading linear term in α :

$$\lim_{n \rightarrow \infty} \mathbb{E}[\theta_n^{(\alpha)}] = \theta^* + \alpha \Delta + O(\alpha^{3/2}),$$

where $\Delta = \bar{A}^{-1} \sum_{k=1}^{\infty} \mathbb{E}[\{\mathbf{Q}^k \tilde{A}(Z_{\infty})\} \epsilon(Z_{\infty})]$ depends on the correlation structure of the Markov chain, and $\tilde{A}(z) = A(z) - \bar{A}$ is the centered matrix-valued function. Under stronger expansion assumptions, the power-series approach of Huo, Chen, and Xie (2023) gives higher-order bias expansions in integer powers of α ; in the Levin decomposition, the first misadjustment bias component itself has an $O(\alpha^2)$ remainder after the leading $\alpha\Delta$ term.

1.2. Richardson–Romberg extrapolation

To eliminate the leading $O(\alpha)$ bias term, we employ the *Richardson–Romberg* (RR) *extrapolation* procedure. Two LSA sequences are run *on the same Markov chain trajectory* $\{Z_k\}$ with step sizes α and 2α , and the RR iterate is formed as

$$\bar{\theta}_n^{(\alpha, \text{RR})} = 2\bar{\theta}_n^{(\alpha)} - \bar{\theta}_n^{(2\alpha)}.$$

Since both sequences share the same noise realization, the leading bias term $\alpha\Delta$ cancels, leaving a residual bias of order $O(\alpha^{3/2})$ or higher (Levin et al., 2025).

More generally, one can consider the multi-level extrapolation with M step sizes $\mathcal{A} = \{\alpha_1, \dots, \alpha_M\}$ and coefficients $\{h_m\}$ determined by the Vandermonde system (Huo et al., 2023):

$$\sum_{m=1}^M h_m = 1, \quad \sum_{m=1}^M h_m \alpha_m^l = 0, \quad l = 1, \dots, M-1,$$

which cancels successive powers in settings where such a power-series expansion is available.

1.3. Problem statement and goals

The high-order moment bounds for the RR iterate $\bar{\theta}_n^{(\alpha, \text{RR})}$ established in Levin et al. (2025) show that the leading error term scales as $\sqrt{\text{Tr } \Sigma_{\epsilon}^{(M)}} \cdot n^{-1/2}$, where $\Sigma_{\epsilon}^{(M)}$ is the Markovian noise covariance defined below in the key quantities subsection, matching the minimax optimal rate. Berry–Esseen type bounds and bootstrap inference procedures for the *standard* Polyak–Ruppert average $\bar{\theta}_n$ (without extrapolation) under Markovian noise have been obtained in Samsonov, Sheshukova, Moulines, and Naumov (2025).

However, the *distributional approximation* — specifically, the central limit theorem and Berry–Esseen bounds — for the *averaged Richardson–Romberg iterates* $\bar{\theta}_n^{(\alpha, \text{RR})}$ remains an open problem. The main goal of this work is to fill this gap by establishing:

1. A *central limit theorem* for $\sqrt{n}(\bar{\theta}_n^{(\alpha, \text{RR})} - \theta^*)$, identifying the limiting covariance matrix.
2. *Non-asymptotic Berry–Esseen bounds* for the rate of convergence to the Gaussian limit.
3. A practical *inference procedure* (confidence intervals) based on the RR-extrapolated iterates.

1.4. Setting and assumptions

We now formalize the setting and state the assumptions that will be used throughout this work.

Let $\{Z_k\}_{k \in \mathbb{N}}$ be a Markov chain on a complete separable metric space $(Z, \mathcal{Z})$ with transition kernel Q .

Assumption 1 (Uniform geometric ergodicity). The kernel Q admits a unique invariant distribution π and is *uniformly geometrically ergodic*: there exists $t_{\text{mix}} \in \mathbb{N}^*$ such that for all $k \in \mathbb{N}^*$,

$$\Delta(Q^k) := \sup_{z, z' \in Z} \frac{1}{2} \|Q^k(z, \cdot) - Q^k(z', \cdot)\|_{\text{TV}} \leq (1/4)^{\lfloor k/t_{\text{mix}} \rfloor}.$$

Equivalently, there exist constants $\zeta > 0$ and $\rho \in (0, 1)$ such that $\sup_z \|Q^k(z, \cdot) - \pi\|_{\text{TV}} \leq \zeta \rho^k$ for all $k \geq 1$.

Assumption 2 (Hurwitz condition and boundedness). The matrix $-\bar{A}$ is Hurwitz, i.e., all eigenvalues of $\bar{A}$ have strictly positive real parts. Moreover,

$$C_A := \max\left(\sup_{z \in Z} \|A(z)\|, \sup_{z \in Z} \|\tilde{A}(z)\|\right) < \infty,$$

where $\tilde{A}(z) := A(z) - \bar{A}$.

Assumption 3 (Noise regularity). The noise function $\epsilon(z) = \tilde{A}(z)\theta^* - \tilde{b}(z)$, where $\tilde{b}(z) = b(z) - \bar{b}$, satisfies

$$\|\epsilon\|_\infty := \sup_{z \in Z} \|\epsilon(z)\| < +\infty.$$

Under Assumptions 1–3, the error $\theta_k^{(\alpha)} - \theta^*$ satisfies the recursion

$$\theta_k^{(\alpha)} - \theta^* = (I - \alpha A(Z_k))(\theta_{k-1}^{(\alpha)} - \theta^*) - \alpha \epsilon(Z_k).$$

1.5. Key quantities

The *Markovian noise covariance matrix* captures both the marginal variance and the temporal correlations of the noise:

$$\Sigma_\epsilon^{(M)} = \mathbb{E}_\pi[\epsilon(Z_0)\epsilon(Z_0)^\top] + \sum_{\ell=1}^{\infty} \left(\mathbb{E}_\pi[\epsilon(Z_0)\epsilon(Z_\ell)^\top] + \mathbb{E}_\pi[\epsilon(Z_\ell)\epsilon(Z_0)^\top] \right).$$

This matrix is the limiting covariance in the Markov chain CLT for the partial sums $n^{-1/2} \sum_{t=0}^{n-1} \epsilon(Z_t)$ (cf. Douc et al., 2018, Theorem 21.2.10).

The *asymptotically optimal covariance matrix* is given by

$$\Sigma_\infty = \bar{A}^{-1} \Sigma_\epsilon^{(M)} (\bar{A}^{-1})^\top.$$

This is the covariance target attained by the averaged linearized recursion. We call it optimal in the usual averaged-SA sense; a full Hájek–Le Cam optimality statement would require an additional local-asymptotic experiment argument, which is not part of this thesis.

The *Lyapunov equation* plays a central role in the contraction analysis. For any $P = P^\top \succ 0$, there exists a unique $Q = Q^\top \succ 0$ satisfying $\bar{A}^\top Q + Q \bar{A} = P$. Defining $a = \lambda_{\min}(P)/(2 \|Q\|)$ and $\kappa_Q = \lambda_{\max}(Q)/\lambda_{\min}(Q)$, the key contraction property holds: for all $\alpha \in [0, \alpha_\infty]$,

$$\|I - \alpha \bar{A}\|_Q^2 \leq 1 - \alpha a.$$

Since the iterates $\theta_k^{(\alpha)}$ alone are generally not Markovian (due to the Markovian noise), we consider the *joint process* $(\theta_k^{(\alpha)}, Z_{k+1})$ with kernel

$$\bar{P}_\alpha f(\theta, z) = \int_Z Q(z, dz') f(F_{z'}(\theta), z'),$$

where $F_z(\theta) = (I - \alpha A(z))\theta + \alpha b(z)$. Under Assumptions 1–3, this joint chain admits a unique invariant distribution Π_α for sufficiently small $\alpha > 0$ (Levin et al., 2025).

2. Zeroth-Order Richardson–Romberg Difference

2.1. LSA Error Decomposition

We consider the recursion

$$\theta_k = \theta_{k-1} - \alpha_k(A(Z_k)\theta_{k-1} - b(Z_k)), \quad \alpha_k = \alpha = \text{const.}$$

Define the transition products

$$\Gamma_{m:k} = \prod_{l=m}^k (I - \alpha A(Z_l)).$$

Write $B_\alpha := I - \alpha \bar{A}$ and introduce the deterministic-product linearized term

$$J_k^{(0,\alpha)} = -\alpha \sum_{l=1}^k B_\alpha^{k-l} \epsilon(Z_l), \quad J_0^{(0,\alpha)} = 0.$$

The difference between the exact random-product expansion and this deterministic-product term is denoted by $R_k^{(\alpha)}$:

$$\theta_k^{(\alpha)} - \theta^* = J_k^{(0,\alpha)} + B_\alpha^k(\theta_0 - \theta^*) + R_k^{(\alpha)}.$$

This convention matches the weight decomposition in Section 4.1.

A standard PR-averaged decomposition (cf. Chapter 4 for the full derivation) yields

$$\sqrt{n}(\bar{\theta}_n^{(\alpha)} - \theta^*) = W + D_1,$$

where the leading martingale-like term is

$$W = -\frac{1}{\sqrt{n}} \sum_{l=1}^{n-1} Q_l \epsilon(Z_l), \quad Q_l = \alpha \sum_{k=l}^{n-1} B_\alpha^{k-l},$$

and the residual term is

$$D_1 = \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} B_\alpha^k(\theta_0 - \theta^*) + \frac{1}{\sqrt{n}} \sum_{k=1}^{n-1} R_k^{(\alpha)}.$$

2.2. RR Combination and the $\tilde{J}_n^{(0,\alpha)}$ Term

Applying the deterministic-product decomposition of the previous subsection separately to step sizes α and 2α , and writing $B_{2\alpha} := I - 2\alpha \bar{A}$, the Richardson–Romberg combination has the form

$$\begin{aligned} \theta_n^{(\text{RR},\alpha)} - \theta^* &= [2B_\alpha^n - B_{2\alpha}^n](\theta_0 - \theta^*) \\ &\quad + [2J_n^{(0,\alpha)} - J_n^{(0,2\alpha)}] + [2R_n^{(\alpha)} - R_n^{(2\alpha)}]. \end{aligned}$$

The first bracket is the deterministic transient in this convention. The second bracket is the linearized stochastic RR difference, and the last one is the higher-order random-product remainder. In this subsection we focus on the zeroth-order RR difference, which we denote

$$\tilde{J}_n^{(0,\alpha)} = 2J_n^{(0,\alpha)} - J_n^{(0,2\alpha)},$$

where, by definition,

$$J_n^{(0,\alpha)} = -\alpha \sum_{j=1}^n (I - \alpha \bar{A})^{n-j} \epsilon(Z_j).$$

Substituting and using the elementary identity $X^m - Y^m = (X - Y) \sum_{i=1}^m X^{i-1} Y^{m-i}$ with $X = I - \alpha \bar{A}$, $Y = I - 2\alpha \bar{A}$, $X - Y = \alpha \bar{A}$, we get

$$\begin{aligned} \tilde{J}_n^{(0,\alpha)} &= -2\alpha \sum_{j=1}^n \left[(I - \alpha \bar{A})^{n-j} - (I - 2\alpha \bar{A})^{n-j} \right] \epsilon(Z_j) \\ &= -2\alpha^2 \bar{A} \sum_{j=1}^n \underbrace{\sum_{i=1}^{n-j} (I - \alpha \bar{A})^{i-1} (I - 2\alpha \bar{A})^{n-j-i}}_{=: H_j^{(n)}} \epsilon(Z_j) \\ &= -2\alpha^2 \bar{A} \sum_{j=1}^n H_j^{(n)} \epsilon(Z_j). \end{aligned}$$

The extra factor $\alpha \bar{A}$ pulled out front is the source of the additional α -decay of the RR difference compared to a single LSA trajectory.

2.3. Norm Estimate for $H_j^{(n)}$

To bound $\|H_j^{(n)}\|$ we use the following standard result.

Lemma 1. Let $-\bar{A}$ be a Hurwitz matrix. Then for any $P = P^\top \succ 0$ there exists a unique $Q = Q^\top \succ 0$ satisfying

$$\bar{A}^\top Q + Q \bar{A} = P.$$

Moreover, letting

$$a := \frac{\lambda_{\min}(P)}{2 \|Q\|}, \quad \alpha_\infty := \min \left(\frac{\lambda_{\min}(P)}{2\kappa_Q \|\bar{A}\|_Q^2}, \frac{\|Q\|}{\lambda_{\min}(P)} \right),$$

with $\kappa_Q := \lambda_{\max}(Q)/\lambda_{\min}(Q)$, one has for all $\alpha \in [0, \alpha_\infty]$:

$$\alpha a \leq 1/2, \quad \|I - \alpha \bar{A}\|_Q^2 \leq 1 - \alpha a.$$

We now estimate $\|H_j^{(n)}\|$ by combining the triangle inequality, submultiplicativity, the equivalence $\|X\| \leq \kappa_Q^{1/2} \|X\|_Q$ (applied to **each** operator-norm factor, hence two factors of $\kappa_Q^{1/2}$ multiplying to κ_Q), and the Lyapunov contraction at step sizes α and 2α :

$$\begin{aligned} \|H_j^{(n)}\| &\leq \sum_{i=1}^{n-j} \|I - \alpha \bar{A}\|^{i-1} \|I - 2\alpha \bar{A}\|^{n-j-i} && \text{(triangle + submult.)} \\ &\leq \kappa_Q \sum_{i=1}^{n-j} (1 - \alpha a)^{(i-1)/2} (1 - 2\alpha a)^{(n-j-i)/2} && \text{(equiv. + Lyapunov)} \\ &= \kappa_Q (1 - \alpha a)^{(n-j-1)/2} \sum_{k=0}^{n-j-1} \left(\frac{1 - 2\alpha a}{1 - \alpha a} \right)^{k/2} && \text{(reindex } k = n - j - i) \\ &\leq \kappa_Q (1 - \alpha a)^{(n-j-1)/2} \frac{1}{1 - \sqrt{\frac{1-2\alpha a}{1-\alpha a}}} && \text{(geometric series).} \end{aligned}$$

For $\alpha a \leq 1/2$ the geometric rate is bounded below by an **elementary** inequality: combining $\sqrt{(1 - 2\alpha a)/(1 - \alpha a)} \leq \sqrt{1 - \alpha a}$ (since $(1 - 2\alpha a) \leq (1 - \alpha a)^2$) with $1 - \sqrt{1 - x} \geq x/2$ on $[0, 1]$ gives

$$1 - \sqrt{\frac{1 - 2\alpha a}{1 - \alpha a}} \geq 1 - \sqrt{1 - \alpha a} \geq \frac{\alpha a}{2}, \quad \text{i.e.} \quad \frac{1}{1 - \sqrt{\frac{1-2\alpha a}{1-\alpha a}}} \leq \frac{2}{\alpha a}.$$

Defining the chapter-local constant

$$\bar{C}_A := \kappa_Q$$

(distinct from the assumption-2 sup-norm constant C_A , which enters separately via $\|\bar{A}\|$ in the next subsection), we arrive at the final estimate

$$\|H_j^{(n)}\| \leq \bar{C}_A (1 - \alpha a)^{(n-j-1)/2} \frac{2}{\alpha a}.$$

The kernel decays geometrically in $n - j$ at rate $\sqrt{1 - \alpha a}$ but its summed weight is of order $1/(\alpha a)$. The next subsection shows how the prefactor α^2 in $\tilde{J}_n^{(0, \alpha)}$ exactly compensates this divergence.

2.4. L^2 Bound for the Zeroth-Order Term

The expression

$$\tilde{J}_n^{(0, \alpha)} = - \sum_{j=1}^n 2\alpha^2 \bar{A} H_j^{(n)} \epsilon(Z_j)$$

is a weighted sum of values of the centered noise function ϵ along the Markov chain. We control its tail via the following Markov concentration statement.

Lemma 2. Assume **UGE 1**. Let $\{g_i\}_{i=1}^n$ be a family of measurable functions $g_i : Z \rightarrow \mathbb{R}^d$ such that

$$c_i = \|g_i\|_\infty < \infty \quad \text{for all } i \geq 1, \quad \pi(g_i) = 0 \quad \text{for all } i \in \{1, \dots, n\}.$$

Then, for any initial distribution ξ on $(Z, \mathcal{Z})$, any $n \in \mathbb{N}$ and any $t \geq 0$,

$$\mathbb{P}_\xi \left(\left\| \sum_{i=1}^n g_i(Z_i) \right\| \geq t \right) \leq 2 \exp \left(- \frac{t^2}{2u_n^2} \right),$$

where

$$u_n = 8 \left(\sum_{i=1}^n c_i^2 \right)^{1/2} \sqrt{t_{\text{mix}}}.$$

Proof. The proof follows the lines of Durmus et al. (2025, Lemma 9).

We apply the lemma with $g_{j(z)} = -2\alpha^2 \bar{A} H_j^{(n)} \epsilon(z)$. Each g_j is centered under π since $\pi(\epsilon) = 0$, so the centering hypothesis holds. Combining the bound $\|\bar{A}\| \leq C_A$ (Assumption 2) with the previous subsection, define

$$\tilde{C}_A := C_A \bar{C}_A = C_A \kappa_Q.$$

Then the per-summand bound is

$$\begin{aligned} \|g_j\|_\infty &\leq 2\alpha^2 \tilde{C}_A (1 - \alpha a)^{(n-j-1)/2} \frac{2}{\alpha a} \|\epsilon\|_\infty \\ &= 4\alpha \tilde{C}_A \|\epsilon\|_\infty (1 - \alpha a)^{(n-j-1)/2}. \end{aligned}$$

Note the prefactor α^2 collapsing to α : the $1/(\alpha a)$ blow-up in $H_j^{(n)}$ is exactly absorbed by one factor of α . Squaring and summing over j , using $1 - \alpha a \geq 1/2$:

$$\begin{aligned}
\sum_{j=1}^n \|g_j\|_\infty^2 &\leq 16\alpha^2 \tilde{C}_A^2 \|\epsilon\|_\infty^2 \sum_{j=1}^n (1-\alpha a)^{n-j-1} \\
&\leq 16\alpha^2 \tilde{C}_A^2 \|\epsilon\|_\infty^2 \frac{1}{(1-\alpha a)\alpha a} \\
&\leq \frac{32\alpha \tilde{C}_A^2 \|\epsilon\|_\infty^2}{a}.
\end{aligned}$$

Plugging this into the variance proxy $u_n^2 = 64 t_{\text{mix}} \sum_{j=1}^n \|g_j\|_\infty^2$ from the lemma and defining

$$\hat{C}_A := \sqrt{2048} \tilde{C}_A \|\epsilon\|_\infty \sqrt{\frac{t_{\text{mix}}}{a}} = 32\sqrt{2} \tilde{C}_A \|\epsilon\|_\infty \sqrt{\frac{t_{\text{mix}}}{a}},$$

gives

$$u_n^2 \leq \hat{C}_A^2 \alpha.$$

To convert the sub-Gaussian tail into a moment bound, we use the following standard fact.

Lemma 3. Let X be an $\mathbb{R}^d$ -valued random variable satisfying

$$\mathbb{P}(\|X\| \geq t) \leq 2 \exp\left(-\frac{t^2}{2\sigma^2}\right) \quad \text{for all } t \geq 0,$$

for some $\sigma^2 > 0$. Then, for any $p \geq 2$, it holds that

$$\mathbb{E}[\|X\|^p] \leq 2 p^{p/2} \sigma^p.$$

Applying the lemma with $\sigma^2 = u_n^2 \leq \hat{C}_A^2 \alpha$ and an arbitrary $p \geq 2$ gives

$$\mathbb{E}^{1/p}[\|\tilde{J}_n^{(0,\alpha)}\|^p] \leq 2^{1/p} \sqrt{p} \hat{C}_A \sqrt{\alpha} \leq 2\sqrt{p} \hat{C}_A \sqrt{\alpha},$$

or equivalently

$$\frac{1}{\sqrt{\alpha}} \mathbb{E}^{1/p}[\|\tilde{J}_n^{(0,\alpha)}\|^p] \leq 2\sqrt{p} \hat{C}_A.$$

The zeroth-order RR difference is therefore $O(\sqrt{\alpha})$ in L^p for every $p \geq 2$, uniformly in n — an order- $\sqrt{\alpha}$ gain over a single LSA trajectory, which is $O(1)$ in L^p in the stationary regime. The $p = 2$ statement used in the rest of the analysis is a special case.

3. Last Iterate Analysis

3.1. Centered Bound for the Shifted First-Order Perturbation

The purpose of this section is to isolate the last-iteration weighted term which appears in the analysis of $J_n^{(1,\alpha)}$, and to record a clean L_p bound for its centered part. The proof uses the deterministic-product perturbation expansion from Samsonov et al. (2025, Proposition 9), specialized to the constant-stepsize setting.

Let

$$B = I - \alpha \bar{A}.$$

Throughout this section we assume $0 < \alpha \leq \alpha_\infty$, so that

$$\|B^m\|_Q \leq (1 - \alpha a)^{m/2}, \quad m \geq 0.$$

All constants denoted by C may depend on C_A , κ_Q , and norm equivalence constants, but not on α , n , p , or t_{mix} .

The deterministic-product components are defined by

$$J_n^{(0,\alpha)} = B J_{n-1}^{(0,\alpha)} - \alpha \epsilon(Z_n), \quad J_0^{(0,\alpha)} = 0,$$

and

$$J_n^{(1,\alpha)} = B J_{n-1}^{(1,\alpha)} - \alpha \tilde{A}(Z_n) J_{n-1}^{(0,\alpha)}, \quad J_0^{(1,\alpha)} = 0.$$

Hence

$$J_n^{(0,\alpha)} = -\alpha \sum_{k=1}^n B^{n-k} \epsilon(Z_k),$$

and

$$\begin{aligned} J_n^{(1,\alpha)} &= -\alpha \sum_{j=1}^n B^{n-j} \tilde{A}(Z_j) J_{j-1}^{(0,\alpha)} \\ &= \alpha^2 \sum_{1 \leq k < j \leq n} B^{n-j} \tilde{A}(Z_j) B^{j-1-k} \epsilon(Z_k) \\ &= \alpha^2 \sum_{k=1}^{n-1} \sum_{r=1}^{n-k} B^{n-k-r} \tilde{A}(Z_{k+r}) B^{r-1} \epsilon(Z_k). \end{aligned}$$

The sign in the second line is positive because $J_{j-1}^{(0,\alpha)}$ already contains the minus sign.

The last display is for $J_n^{(1,\alpha)}$ itself. The last-iteration estimate below operates on a **shifted** version, obtained by inserting one additional left factor B into the recursion:

$$S_n = \sum_{t=0}^{n-1} B^{n-t} \tilde{A}(Z_{t+1}) J_t^{(0,\alpha)}.$$

Reindexing $j = t + 1$ rewrites S_n in the same form as $J_n^{(1,\alpha)}$ but with one extra power of B :

$$S_n = \sum_{j=1}^n B^{n-j+1} \tilde{A}(Z_j) J_{j-1}^{(0,\alpha)} = -\frac{1}{\alpha} B J_n^{(1,\alpha)}.$$

Thus the corresponding shifted first-order contribution is

$$T_n^{(1,\alpha)} = -\alpha S_n = B J_n^{(1,\alpha)},$$

i.e. $T_n^{(1,\alpha)}$ is exactly $J_n^{(1,\alpha)}$ pre-multiplied by one additional B . Transferring a bound from $T_n^{(1,\alpha)}$ back to $J_n^{(1,\alpha)}$ therefore requires a local inverse bound on $B^{-1} = (I - \alpha \bar{A})^{-1}$; this inverse-bound step is used explicitly when the shifted estimate is applied below.

Lemma 4. Assume the Markov chain is started from stationarity, that is, the law of Z_1 is π , and $\pi(\tilde{A}) = 0$. For every $p \geq 2$,

$$\|S_n - \mathbb{E}S_n\|_{L_p} \leq C \|\epsilon\|_\infty \left(p^{3/2} t_{\text{mix}}^{1/2} \frac{1}{a} + p^{1/2} t_{\text{mix}}^{3/2} \sqrt{\frac{\alpha}{a}} \right).$$

Consequently,

$$\|T_n^{(1,\alpha)} - \mathbb{E}T_n^{(1,\alpha)}\|_{L_p} \leq C\alpha \|\epsilon\|_\infty \left(p^{3/2} t_{\text{mix}}^{1/2} \frac{1}{a} + p^{1/2} t_{\text{mix}}^{3/2} \sqrt{\frac{\alpha}{a}} \right).$$

Proof. Since $J_0^{(0,\alpha)} = 0$, the term $t = 0$ in S_n vanishes. Substituting the explicit formula for $J_t^{(0,\alpha)}$ gives

$$\begin{aligned} S_n &= -\alpha \sum_{t=1}^{n-1} \sum_{k=1}^t B^{n-t} \tilde{A}(Z_{t+1}) B^{t-k} \epsilon(Z_k) \\ &= -\alpha \sum_{k=1}^{n-1} H_{k+1}^{(w)} \epsilon(Z_k), \end{aligned}$$

where, after the change of summation index $l = t - k + 1$,

$$H_{k+1}^{(w)} = \sum_{l=1}^{n-k} B^{n-k-l+1} \tilde{A}(Z_{k+l}) B^{l-1}.$$

The kernel $H_{k+1}^{(w)}$ acts on the past noise $\epsilon(Z_k)$ through future states Z_{k+l} , so $H_{k+1}^{(w)} \epsilon(Z_k)$ is a future-weighted bilinear functional of the trajectory.

Define

$$\mu_k^{(w)} = \mathbb{E}_\pi \left[H_{k+1}^{(w)} \epsilon(Z_k) \right].$$

Then

$$S_n - \mathbb{E}S_n = -\alpha \sum_{k=1}^{n-1} \left\{ H_{k+1}^{(w)} \epsilon(Z_k) - \mu_k^{(w)} \right\}.$$

Let $\mathcal{F}_k = \sigma(Z_1, \dots, Z_k)$. The Markov property gives

$$\mathbb{E} \left[H_{k+1}^{(w)} \epsilon(Z_k) \mid \mathcal{F}_k \right] = v_k^{(w,\varepsilon)}(Z_k),$$

where

$$v_k^{(w)}(z) = \sum_{l=1}^{n-k} B^{n-k-l+1} (Q^l \tilde{A})(z) B^{l-1}, \quad v_k^{(w,\varepsilon)}(z) = v_k^{(w)}(z) \epsilon(z).$$

Here Q denotes the one-step Markov transition kernel of $(Z_k)_{k \geq 1}$, acting on bounded matrix-valued functions by integration against the conditional law:

$$(Q\tilde{A})(z) = \int \tilde{A}(u) Q(z, du) = \mathbb{E}[\tilde{A}(Z_{k+1}) \mid Z_k = z],$$

and Q^l is its l -fold iterate, the l -step kernel $Q^l(z, du) = \mathbb{P}(Z_{k+l} \in du \mid Z_k = z)$, so that

$$(Q^l \tilde{A})(z) = \int \tilde{A}(u) Q^l(z, du) = \mathbb{E}[\tilde{A}(Z_{k+l}) \mid Z_k = z].$$

In particular $(Q^l \tilde{A})(z) \rightarrow \pi(\tilde{A}) = 0$ at the geometric rate dictated by UGE, which is the only fact about Q^l used below.

Under stationarity, $\mu_k^{(w)} = \pi(v_k^{(w, \varepsilon)})$. Therefore

$$S_n - \mathbb{E}S_n = -\alpha(U_M + U_R),$$

with

$$U_M = \sum_{k=1}^{n-1} \left\{ H_{k+1}^{(w)} \epsilon(Z_k) - v_k^{(w, \varepsilon)}(Z_k) \right\}$$

and

$$U_R = \sum_{k=1}^{n-1} \left\{ v_k^{(w, \varepsilon)}(Z_k) - \pi(v_k^{(w, \varepsilon)}) \right\}.$$

This decomposition splits the centered statistic into two structurally different parts: U_R is an ordinary centered additive functional of the original Markov chain, while U_M is a future-centered bilinear term — the conditional expectation given $\mathcal{F}_k$ vanishes summand-wise, but the summands are not forward martingale differences. We bound each piece separately.

Step 1: bound on U_R . Because $\pi(\tilde{A}) = 0$, applying Q^l followed by integration against π replaces $\tilde{A}$ by its l -step propagation away from stationarity. UGE then gives the geometric Dobrushin bound

$$\|(Q^l \tilde{A})(z)\| = \left\| \int \tilde{A}(u) \{Q^l(z, du) - \pi(du)\} \right\| \leq 2C_A \Delta(Q^l) \leq 2C_A (1/4)^{\lfloor l/t_{\text{mix}} \rfloor}.$$

Inserting this into $v_k^{(w)}$ and factoring out the slow rate $(1 - \alpha a)^{(n-k)/2}$ from the two B -powers, the inner sum becomes a fast-decaying l -series:

$$\begin{aligned} \|v_k^{(w)}\|_\infty &\leq C \sum_{l=1}^{n-k} (1 - \alpha a)^{(n-k-l+1)/2} \Delta(Q^l) (1 - \alpha a)^{(l-1)/2} \quad (\text{triangle} + \text{Lyapunov}) \\ &\leq C(1 - \alpha a)^{(n-k)/2} \sum_{l=1}^{\infty} (1/4)^{\lfloor l/t_{\text{mix}} \rfloor} \quad (\text{extend to } l = \infty) \\ &\leq C t_{\text{mix}} (1 - \alpha a)^{(n-k)/2} \quad (\text{geometric block sum}). \end{aligned}$$

Multiplying by $\|\epsilon\|_\infty$ gives

$$\|v_k^{(w, \varepsilon)}\|_\infty \leq C t_{\text{mix}} \|\epsilon\|_\infty (1 - \alpha a)^{(n-k)/2}.$$

The functions $v_k^{(w, \varepsilon)}(Z_k) - \pi(v_k^{(w, \varepsilon)})$ are centered under π and uniformly bounded by the previous display, so the weighted Markov concentration/Rosenthal bound for centered time-dependent functions yields

$$\begin{aligned}
\|U_R\|_{L_p} &\leq Cp^{1/2}t_{\text{mix}}^{1/2} \left(\sum_{k=1}^{n-1} \|v_k^{(w,\varepsilon)}\|_{\infty}^2 \right)^{1/2} && \text{(Rosenthal)} \\
&\leq Cp^{1/2}t_{\text{mix}}^{3/2} \|\epsilon\|_{\infty} \left(\sum_{k=1}^{n-1} (1-\alpha a)^{n-k} \right)^{1/2} && \text{(plug previous bound)} \\
&\leq Cp^{1/2}t_{\text{mix}}^{3/2} \|\epsilon\|_{\infty} \frac{1}{\sqrt{\alpha a}} && \text{(geometric series).}
\end{aligned}$$

Step 2: bound on U_M . Project onto a deterministic unit vector u and unfold $H_{k+1}^{(w)}u$ as a sum over future states:

$$H_{k+1}^{(w)}u = \sum_{l=1}^{n-k} g_{k,l}(Z_{k+l}), \quad g_{k,l}(z) = B^{n-k-l+1} \tilde{A}(z) B^{l-1} u.$$

Each $g_{k,l}$ is centered under π (since $\pi(\tilde{A}) = 0$), and the two B -powers give the uniform bound

$$\|g_{k,l}\|_{\infty} \leq C(1-\alpha a)^{(n-k)/2},$$

which is independent of l . Squaring and summing over l produces only linear growth in $n-k$:

$$\sum_{l=1}^{n-k} \|g_{k,l}\|_{\infty}^2 \leq C(n-k)(1-\alpha a)^{n-k}.$$

Note the contrast with the U_R analysis: there, UGE folded the l -sum into a single t_{mix} factor, whereas here the same l -independent bound is applied $(n-k)$ times — the price of conditional centering being weaker than π -centering.

Applying the Markov concentration lemma to the future chain conditionally on $\mathcal{F}_k$,

$$\|H_{k+1}^{(w)}u - \mathbb{E}[H_{k+1}^{(w)}u \mid \mathcal{F}_k]\|_{L_p} \leq Cp^{1/2}t_{\text{mix}}^{1/2} \sqrt{(n-k)(1-\alpha a)^{n-k}},$$

and multiplying by $\epsilon(Z_k)$ (which is $\mathcal{F}_k$ -measurable and bounded by $\|\epsilon\|_{\infty}$) gives

$$\|H_{k+1}^{(w)}\epsilon(Z_k) - v_k^{(w,\varepsilon)}(Z_k)\|_{L_p} \leq Cp^{1/2}t_{\text{mix}}^{1/2} \|\epsilon\|_{\infty} \sqrt{(n-k)(1-\alpha a)^{n-k}}.$$

The summands in U_M are centered conditionally on $\mathcal{F}_k$, but they are *not* forward martingale differences: the kernel $H_{k+1}^{(w)}$ peeks into the future of Z_k . To assemble the per- k bounds into a bound on the sum we therefore invoke the weighted Burkholder/block-coupling estimate for future-centered bilinear Markov sums of Samsonov et al. (2025, Proposition 9), specialized to the present constant-stepsizes kernel:

$$\|U_M\|_{L_p} \leq Cp \left(\sum_{k=1}^{n-1} \|H_{k+1}^{(w)}\epsilon(Z_k) - v_k^{(w,\varepsilon)}(Z_k)\|_{L_p}^2 \right)^{1/2}.$$

Plugging in the previous bound and evaluating the resulting sum:

$$\begin{aligned}
\|U_M\|_{L_p} &\leq Cp^{3/2}t_{\text{mix}}^{1/2} \|\epsilon\|_{\infty} \left(\sum_{k=1}^{n-1} (n-k)(1-\alpha a)^{n-k} \right)^{1/2} && \text{(plug previous bound)} \\
&\leq Cp^{3/2}t_{\text{mix}}^{1/2} \|\epsilon\|_{\infty} \frac{1}{\alpha a} && \left(\sum_m m r^m \leq C(1-r)^{-2}, r = 1-\alpha a \right).
\end{aligned}$$

The extra factor $1/\sqrt{\alpha a}$ relative to U_R is precisely the cost of using conditional rather than stationary centering.

Step 3: assembly. Combining the two pieces via $S_n - \mathbb{E}S_n = -\alpha(U_M + U_R)$ and the triangle inequality,

$$\begin{aligned} \|S_n - \mathbb{E}S_n\|_{L_p} &\leq \alpha (\|U_M\|_{L_p} + \|U_R\|_{L_p}) \\ &\leq C \|\epsilon\|_\infty \left(p^{3/2} t_{\text{mix}}^{1/2} \frac{1}{a} + p^{1/2} t_{\text{mix}}^{3/2} \sqrt{\frac{\alpha}{a}} \right). \end{aligned}$$

The first term (from U_M) is the leading contribution for small α ; the second (from U_R) carries the heavier t_{mix} dependence but vanishes as $\alpha \rightarrow 0$. Multiplying by α gives the asserted bound for $T_n^{(1,\alpha)} = -\alpha S_n$. $\square$

Remark 1. The stationarity assumption is used only to avoid boundary notation in this local calculation. For a non-stationary initial law, UGE gives the same bulk estimate plus the standard exponentially small mixing boundary terms. The bias of $T_n^{(1,\alpha)}$ is a separate issue and is controlled through the stationary expansion of $\mathbb{E}J_n^{(1,\alpha)}$ in Levin et al. (2025).

3.2. Application to the PR-averaged RR Misadjustment

The PR-averaged Richardson–Romberg expansion produces, after Step (S8) of the Samsonov scheme applied separately at step sizes α and 2α , a “misadjustment” remainder

$$D_1^{\text{mis, RR}} = \frac{1}{\sqrt{n}} \sum_{k=n_0}^{n-1} \left(2J_k^{(1,\alpha)} - J_k^{(1,2\alpha)} \right),$$

whose centered part must be controlled to feed into a Berry–Esseen statement for $\sqrt{n}(\bar{\theta}_n^{(\text{RR},\alpha)} - \theta^*)$. The bias is not the obstruction: by Levin et al. (2025, Proposition 2),

$$\mathbb{E}_\pi [J_\infty^{(1,\alpha)}] = \alpha\Delta + O(\alpha^2),$$

so the linear term $\alpha\Delta$ cancels in the RR-combination and the stationary bias of $D_1^{\text{mis, RR}}$ is $O(\alpha^2)$. What remains is the centered fluctuation.

Define

$$\Phi(p, \alpha) := p^{3/2} \frac{t_{\text{mix}}^{1/2}}{a} + p^{1/2} t_{\text{mix}}^{3/2} \sqrt{\alpha/a}.$$

The lemma applied at α and at 2α gives, separately,

$$\|T_n^{(1,\alpha)} - \mathbb{E}T_n^{(1,\alpha)}\|_{L_p} \leq C\alpha \Phi(p, \alpha), \quad \|T_n^{(1,2\alpha)} - \mathbb{E}T_n^{(1,2\alpha)}\|_{L_p} \leq C\alpha \Phi(p, 2\alpha) \leq C'\alpha \Phi(p, \alpha),$$

where $C' = \sqrt{2}C$ absorbs the $\sqrt{2}$ -factor coming from the 2α scaling. Combining the two by the triangle inequality and using the index-shift identity $T_k^{(1,w)} = (I - w\bar{A}) J_k^{(1,w)}$ together with the local inverse bound $\|(I - w\bar{A})^{-1}\| \leq 2$ for $w \in \{\alpha, 2\alpha\}$, we get

$$\|(2J_k^{(1,\alpha)} - J_k^{(1,2\alpha)}) - \mathbb{E}(2J_k^{(1,\alpha)} - J_k^{(1,2\alpha)})\|_{L_p} \leq C\alpha \Phi(p, \alpha),$$

uniformly in k . PR-averaging through $\frac{\sqrt{n}}{n-n_0}$ and absorbing the constant therefore yields

$$\|D_1^{\text{mis, RR}}\|_{L_p} = O(\sqrt{n}\alpha).$$

At the optimal scale $\alpha \asymp n^{-1/2}$ this is $O(1)$, and as a remainder in a Berry–Esseen statement of order $n^{-1/4}$ it is **useless**: the misadjustment must be at least $o(1)$ to be subleading.

3.3. Richardson–Romberg Difference of S_n

The single- α centered bound, applied separately at α and 2α and combined termwise via $\|2X - Y\| \leq 2\|X\| + \|Y\|$, produces a misadjustment estimate of order $O(\sqrt{n}\alpha) = O(1)$ at $\alpha \asymp n^{-1/2}$. This estimate

is **blind to coupling**: it never uses the fact that $S_n^{(\alpha)}$ and $S_n^{(2\alpha)}$ share the same noise realization $\{Z_k\}$, and is the one that holds for a generic centred difference of two centred quantities. The Richardson–Romberg construction nonetheless promises an additional factor of α from the deterministic prefactors, via the elementary identity $X^m - Y^m = (X - Y) \sum_{i=1}^m X^{i-1} Y^{m-i}$ at $X = B_\alpha$, $Y = B_{2\alpha}$, $X - Y = \alpha \bar{A}$.

We accordingly view the Richardson–Romberg difference of $S_n^{(\alpha)}$ as a single weighted sum on the common noise, with a deterministic RR-kernel in place of the single- α kernel $H^{(w)}$. The structure of the analysis matches the previous subsection — single-kernel form, conditional/stationary centering, Markov-Rosenthal and weighted Burkholder — with one new ingredient, a pointwise bound on the RR-kernel that uses the identity above to extract a factor of α uniformly across the kernel.

3.3.1. Single-Kernel Form

For step sizes α and 2α , write $B_\alpha = I - \alpha \bar{A}$, $B_{2\alpha} = I - 2\alpha \bar{A}$, and consider the **Richardson–Romberg difference** of the shifted statistic at fixed horizon n :

$$\tilde{S}_n^{\text{RR}} := 2 S_n^{(\alpha)} - S_n^{(2\alpha)}.$$

Both $S_n^{(\alpha)}$ and $S_n^{(2\alpha)}$ are driven by the **same** noise realization $\{Z_k\}$. Substituting $S_n^{(\alpha)} = -\alpha \sum_{k=1}^{n-1} H_{k+1}^{(w),\alpha} \epsilon(Z_k)$ from the previous subsection at both step sizes and combining,

$$\tilde{S}_n^{\text{RR}} = -2\alpha \sum_{k=1}^{n-1} \left[H_{k+1}^{(w),\alpha} - H_{k+1}^{(w),2\alpha} \right] \epsilon(Z_k) = - \sum_{k=1}^{n-1} \mathcal{H}_{k+1}^{\text{RR}} \epsilon(Z_k),$$

with the **RR kernel**

$$\mathcal{H}_{k+1}^{\text{RR}} := 2\alpha \sum_{l=1}^{n-k} \left[B_\alpha^{a_l} \tilde{A}(Z_{k+l}) B_\alpha^{b_l} - B_{2\alpha}^{a_l} \tilde{A}(Z_{k+l}) B_{2\alpha}^{b_l} \right], \quad a_l := n - k - l + 1, \quad b_l := l - 1.$$

By construction $a_l + b_l = n - k$. The future-state factor $\tilde{A}(Z_{k+l})$ enters $\mathcal{H}^{\text{RR}}$ in exactly the same way as in $H^{(w)}$: only the deterministic prefactors $B_\alpha^{a_l}, B_{2\alpha}^{a_l}$ carry the RR cancellation.

3.3.2. Pointwise Kernel Bound

Lemma 5. Let $\alpha, 2\alpha \in (0, \alpha_\infty]$, set $\bar{C}_{\text{RR}} := \kappa_Q^{3/2} \|\bar{A}\|$, and let $M \in \mathbb{R}^{d \times d}$ be any deterministic matrix. For all integers $j, m \geq 0$ with $j + m \geq 1$,

$$\| B_\alpha^j M B_\alpha^m - B_{2\alpha}^j M B_{2\alpha}^m \| \leq \bar{C}_{\text{RR}} \alpha (j + m) (1 - \alpha a)^{(j+m-1)/2} \| M \|.$$

Proof. Apply the additive split

$$B_\alpha^j M B_\alpha^m - B_{2\alpha}^j M B_{2\alpha}^m = (B_\alpha^j - B_{2\alpha}^j) M B_\alpha^m + B_{2\alpha}^j M (B_\alpha^m - B_{2\alpha}^m),$$

and the elementary telescope $X^r - Y^r = \sum_{i=1}^r X^{i-1} (X - Y) Y^{r-i}$ with $X = B_\alpha$, $Y = B_{2\alpha}$, $X - Y = \alpha \bar{A}$. For the first piece, submultiplicativity in $\|\cdot\|_Q$ followed by Lyapunov contraction at both step sizes (with $1 - 2\alpha a \leq 1 - \alpha a$) gives

$$\| B_\alpha^j - B_{2\alpha}^j \|_Q \leq \alpha \|\bar{A}\|_Q j (1 - \alpha a)^{(j-1)/2},$$

hence after one application of the equivalence $\|\cdot\| \leq \kappa_Q^{1/2} \|\cdot\|_Q$,

$$\| (B_\alpha^j - B_{2\alpha}^j) M B_\alpha^m \| \leq \alpha \kappa_Q^{3/2} \|\bar{A}\| j (1 - \alpha a)^{(j+m-1)/2} \| M \|.$$

The second piece is symmetric and gives the same bound with j replaced by m . Adding the two yields the claim. $\square$

The lemma is the **only** place where Richardson–Romberg cancellation is used. Two structural features deserve emphasis:

- The factor α extracted by the telescope is the deterministic manifestation of RR coupling: it is the same identity that powered the zeroth-order bound on $\tilde{J}_n^{(0,\alpha)}$ in Chapter 2 and the discrete-derivative bound on $\mathcal{Q}_{l+1}^{\text{RR}} - \mathcal{Q}_l^{\text{RR}}$ in Chapter 4.
- The factor $j + m$ is the count of telescope summands. It cannot be removed without strengthening the geometric decay, which is already optimal for B_α . In the present application $j + m = n - k$ is the **full lag**, and this growth is what spoils the gain in the variance proxies below.

3.3.3. Centered Decomposition

Following the single- α scheme, fix $\mathcal{F}_k = \sigma(Z_1, \dots, Z_k)$. Under stationarity (Z_1 drawn from π , $\pi(\tilde{A}) = 0$), the Markov property gives

$$\mathbb{E}[\mathcal{H}_{k+1}^{\text{RR}} \epsilon(Z_k) \mid \mathcal{F}_k] = v_k^{(w, \text{RR}, \epsilon)}(Z_k),$$

with

$$v_k^{(w, \text{RR})}(z) := 2\alpha \sum_{l=1}^{n-k} [B_\alpha^{a_l}(Q^l \tilde{A})(z) B_\alpha^{b_l} - B_{2\alpha}^{a_l}(Q^l \tilde{A})(z) B_{2\alpha}^{b_l}], \quad v_k^{(w, \text{RR}, \epsilon)}(z) := v_k^{(w, \text{RR})}(z) \epsilon(z),$$

and $\mathbb{E}[v_k^{(w, \text{RR}, \epsilon)}(Z_k)] = \pi(v_k^{(w, \text{RR}, \epsilon)})$. Therefore

$$\tilde{S}_n^{\text{RR}} - \mathbb{E}\tilde{S}_n^{\text{RR}} = -(U_M^{\text{RR}} + U_R^{\text{RR}}),$$

$$U_M^{\text{RR}} = \sum_{k=1}^{n-1} \left\{ \mathcal{H}_{k+1}^{\text{RR}} \epsilon(Z_k) - v_k^{(w, \text{RR}, \epsilon)}(Z_k) \right\}, \quad U_R^{\text{RR}} = \sum_{k=1}^{n-1} \left\{ v_k^{(w, \text{RR}, \epsilon)}(Z_k) - \pi(v_k^{(w, \text{RR}, \epsilon)}) \right\}.$$

The decomposition has the same conditional/stationary split as in the single- α proof; the RR-kernel bound feeds into the variance proxies of each piece.

3.3.4. L_p Bound

Lemma 6. Assume the Markov chain is started from stationarity and $\pi(\tilde{A}) = 0$. For every $p \geq 2$,

$$\|U_R^{\text{RR}}\|_{L_p} \leq C \|\epsilon\|_\infty p^{1/2} t_{\text{mix}}^{3/2} \sqrt{\alpha}/a^{3/2},$$

$$\|U_M^{\text{RR}}\|_{L_p} \leq C \|\epsilon\|_\infty p^{3/2} t_{\text{mix}}^{1/2}/a^2,$$

with a constant C depending only on $C_A, \kappa_Q, \|\bar{A}\|$. Consequently,

$$\|\tilde{S}_n^{\text{RR}} - \mathbb{E}\tilde{S}_n^{\text{RR}}\|_{L_p} \leq C \|\epsilon\|_\infty \left(p^{3/2} t_{\text{mix}}^{1/2}/a^2 + p^{1/2} t_{\text{mix}}^{3/2} \sqrt{\alpha}/a^{3/2} \right).$$

Proof. The structure mirrors the single- α proof; only the variance proxies change. We trace each piece.

Step 1 (U_R^{RR}). By UGE applied to the inner $Q^l \tilde{A}$ factor, $\|(Q^l \tilde{A})(z)\| \leq 2C_A (1/4)^{\lfloor l/t_{\text{mix}} \rfloor}$. Combining with the kernel-difference Lemma (with $j + m = n - k$ uniform in l , and $\|(Q^l \tilde{A})(z)\|$ playing the role of $\|M\|$),

$$\begin{aligned} \|v_k^{(w, \text{RR})}(z)\| &\leq 2\alpha \sum_{l=1}^{n-k} \bar{C}_{\text{RR}} \alpha (n-k) (1-\alpha\alpha)^{(n-k-1)/2} 2C_A (1/4)^{\lfloor l/t_{\text{mix}} \rfloor} (\text{kernel} + \text{UGE}) \\ &\leq C t_{\text{mix}} \alpha^2 (n-k) (1-\alpha\alpha)^{(n-k-1)/2} \quad (\text{geometric block sum}). \end{aligned}$$

Multiplying by $\|\epsilon\|_\infty$,

$$\|v_k^{(w, \text{RR}, \epsilon)}\|_\infty \leq C t_{\text{mix}} \|\epsilon\|_\infty \alpha^2 (n-k) (1-\alpha\alpha)^{(n-k-1)/2}.$$

Squaring and summing over k , with $\sum_{m \geq 1} m^2 (1 - \alpha a)^{m-1} \leq 2/(\alpha a)^3$,

$$\sum_{k=1}^{n-1} \|v_k^{(w, \text{RR}, \epsilon)}\|_\infty^2 \leq C^2 t_{\text{mix}}^2 \|\epsilon\|_\infty^2 \alpha^4 \frac{2}{(\alpha a)^3} = C' t_{\text{mix}}^2 \|\epsilon\|_\infty^2 \alpha / a^3.$$

The weighted Markov concentration of Levin et al. (2025, Lemma 11) applied to the centered functions $v_k^{(w, \text{RR}, \epsilon)}(Z_k) - \pi(v_k^{(w, \text{RR}, \epsilon)})$ then gives

$$\|U_R^{\text{RR}}\|_{L_p} \leq C p^{1/2} t_{\text{mix}}^{1/2} \left(\sum_{k=1}^{n-1} \|v_k^{(w, \text{RR}, \epsilon)}\|_\infty^2 \right)^{1/2} \leq C p^{1/2} t_{\text{mix}}^{3/2} \|\epsilon\|_\infty \sqrt{\alpha} / a^{3/2}.$$

Step 2 (U_M^{RR}). Project onto a unit vector u and unfold the kernel as a future-state sum,

$$\mathcal{H}_{k+1}^{\text{RR}} u = \sum_{l=1}^{n-k} F_l^{\text{RR}}(Z_{k+l}), \quad F_l^{\text{RR}}(z) := 2\alpha [B_{\alpha^l}^{a_l} \tilde{A}(z) B_{\alpha^l}^{b_l} - B_{2\alpha}^{a_l} \tilde{A}(z) B_{2\alpha}^{b_l}] u.$$

Each F_l^{RR} is centered under π and the kernel-difference Lemma gives the **l -uniform** bound

$$\|F_l^{\text{RR}}\|_\infty \leq 2\alpha \bar{C}_{\text{RR}} \alpha (a_l + b_l) (1 - \alpha a)^{(n-k-1)/2} \|\tilde{A}\| \leq C \alpha^2 (n-k) (1 - \alpha a)^{(n-k-1)/2}.$$

Squaring and summing over the $n-k$ values of l ,

$$\sum_{l=1}^{n-k} \|F_l^{\text{RR}}\|_\infty^2 \leq C^2 \alpha^4 (n-k)^3 (1 - \alpha a)^{n-k-1}.$$

Markov concentration on the future chain conditionally on $\mathcal{F}_k$ then yields

$$\|\mathcal{H}_{k+1}^{\text{RR}} u - \mathbb{E}[\mathcal{H}_{k+1}^{\text{RR}} u | \mathcal{F}_k]\|_{L_p, Z_k} \leq C p^{1/2} t_{\text{mix}}^{1/2} \alpha^2 (n-k)^{3/2} (1 - \alpha a)^{(n-k-1)/2},$$

and multiplying by $\|\epsilon(Z_k)\| \leq \|\epsilon\|_\infty$ preserves the bound. Summing the squares over k , with $\sum_{m \geq 1} m^3 (1 - \alpha a)^{m-1} \leq 6/(\alpha a)^4$,

$$\sum_{k=1}^{n-1} \|\mathcal{H}_{k+1}^{\text{RR}} \epsilon(Z_k) - v_k^{(w, \text{RR}, \epsilon)}(Z_k)\|_{L_p}^2 \leq C p t_{\text{mix}} \|\epsilon\|_\infty^2 \alpha^4 / (\alpha a)^4 = C p t_{\text{mix}} \|\epsilon\|_\infty^2 / a^4.$$

The weighted Burkholder estimate for future-centered bilinear Markov sums (Samsonov et al., 2025, Proposition 9; same instance as in the single- α proof) closes the bound

$$\|U_M^{\text{RR}}\|_{L_p} \leq C p \left(\sum_{k=1}^{n-1} \|\mathcal{H}_{k+1}^{\text{RR}} \epsilon(Z_k) - v_k^{(w, \text{RR}, \epsilon)}(Z_k)\|_{L_p}^2 \right)^{1/2} \leq C p^{3/2} t_{\text{mix}}^{1/2} \|\epsilon\|_\infty / a^2.$$

Assembly. Adding the two bounds gives the claim. $\square$

3.3.5. Two Approaches: Triangle Inequality vs. Kernel Difference

There are two natural ways to bound the centered RR-difference, and it is instructive to lay them side by side.

Approach A: triangle inequality on the single- α lemma. The single- α centered bound from the previous subsection,

$$\|S_n^{(\alpha)} - \mathbb{E} S_n^{(\alpha)}\|_{L_p} \leq C \|\epsilon\|_\infty \left(p^{3/2} t_{\text{mix}}^{1/2} / a + p^{1/2} t_{\text{mix}}^{3/2} \sqrt{\alpha/a} \right),$$

applied separately at α and 2α and combined by $\|2X - Y\| \leq 2\|X\| + \|Y\|$, yields

$$\| \tilde{S}_n^{\text{RR}} - \mathbb{E} \tilde{S}_n^{\text{RR}} \|_{L_p} \leq C \| \epsilon \|_{\infty} \left(p^{3/2} t_{\text{mix}}^{1/2} / a + p^{1/2} t_{\text{mix}}^{3/2} \sqrt{\alpha/a} \right).$$

This bound is **blind to coupling**: the same-noise structure of $S_n^{(\alpha)}$ and $S_n^{(2\alpha)}$ does not enter, and the resulting estimate is the one that holds for a generic centred difference of two centred statistics.

Approach B: kernel-difference identity. Express $\tilde{S}_n^{\text{RR}}$ as a single noise-weighted sum with kernel $\mathcal{H}^{\text{RR}}$ and feed the kernel-difference Lemma through the conditional/stationary centring scheme. The resulting bound, derived above, is

$$\| \tilde{S}_n^{\text{RR}} - \mathbb{E} \tilde{S}_n^{\text{RR}} \|_{L_p} \leq C \| \epsilon \|_{\infty} \left(p^{3/2} t_{\text{mix}}^{1/2} / a^2 + p^{1/2} t_{\text{mix}}^{3/2} \sqrt{\alpha} / a^{3/2} \right),$$

i.e. the same dependence on α but an **additional** factor of $1/a$ in both terms. For $a < 1$ — the regime of interest — Approach B is **strictly worse** than Approach A.

The mechanism is structural. The kernel-difference identity extracts a factor of α on each summand of $\mathcal{H}^{\text{RR}}$, but at the price of a count $a_l + b_l = n - k$ of summands carrying that gain **uniformly in l** . Squaring and summing over l in the Markov-Rosenthal/Burkholder variance proxy, the $(n - k)$ factor combines with $\sum (n - k)^q (1 - \alpha a)^{n-k-1} \leq q! / (\alpha a)^{q+1}$ to introduce one extra $1/(\alpha a)$, which is not absorbed by the α^2 kernel prefactor.

The same phenomenon, with weaker symptoms, was already visible in the zeroth-order analysis of Chapter 2: the bound $\| H_j^{(n)} \| \leq \bar{C}_A (1 - \alpha a)^{(n-j-1)/2} 2/(\alpha a)$ has the same telescope-times-geometric-tail structure, and the $1/(\alpha a)$ blow-up there is absorbed by **one** of the two α -factors of the prefactor α^2 — leaving a residual $1/a$ in the variance proxy.

In short, the kernel-difference identity captures only the **deterministic** part of the RR cancellation, through $B_\alpha^m - B_{2\alpha}^m$. It does not see the stochastic coupling through $\{Z_k\}$, and the latter is what the RR construction is for. The kernel-difference route therefore replaces a coupling-blind triangle estimate by an equally coupling-blind kernel estimate, with no net gain on the centered L_p norm of $\tilde{S}_n^{\text{RR}}$.

3.3.6. Implication for the PR-Averaged Misadjustment

Both approaches feed into the misadjustment in the same way. After $\sqrt{n}$ -scaled PR-averaging,

$$\| D_1^{\text{mis, RR}} \|_{L_p} = O(\sqrt{n} \alpha) = O(1) \quad \text{at} \quad \alpha \asymp n^{-1/2},$$

which fails to vanish and does not match the martingale Berry–Esseen scale $n^{-1/4}$. The RR construction provides **no improvement** at this level of analysis: the rate is the same as for the single- α misadjustment.

A bound of order $n^{-1/4}$ on the misadjustment is nevertheless reachable, but through a **different** decomposition. The depth-two perturbation expansion

$$H^{(0,\alpha)} = J^{(1,\alpha)} + J^{(2,\alpha)} + H^{(2,\alpha)}$$

of Levin et al. (2025) splits the misadjustment into three independently controlled pieces:

- the centered bilinear statistic $\sum_t \tilde{A}(Z_{t+1}) J_t^{(0,\alpha)}$ is bounded by Levin’s Proposition 3 / Corollary 6 at order $\sqrt{\alpha r} + 1/\sqrt{\alpha}$ in L_p , contributing $O(n^{-1/4})$ after $\sqrt{n}$ -scaled PR-averaging;
- the high-order moments $\| J_n^{(2,\alpha)} \|_{L_p}$ and $\| H_n^{(2,\alpha)} \|_{L_p}$ are $O(\alpha^{3/2})$ by Levin’s Propositions 8–9, giving the same $O(n^{-1/4})$ contribution after PR-averaging and $\sqrt{n}$ -scaling;
- the bias of $J^{(1,\alpha)}$ has the form $\alpha \Delta + O(\alpha^2)$ by Levin’s Proposition 2, and the leading $\alpha \Delta$ cancels under RR, leaving a stationary bias of order $\alpha^2 = O(n^{-1})$, well below the target.

Stitching these pieces together yields $\| D_1^{\text{mis, RR}} \|_{L_p} = O(n^{-1/4})$. The kernel-difference Lemma proved earlier in this section is **not** a load-bearing ingredient for that rate.

A **strict** improvement of the misadjustment to $O(n^{-1/2})$, subleading to martingale Berry–Esseen, would require a tool that exploits the stochastic coupling of the two trajectories — for instance, Berbee-style decoupling on the joint two-chain state space, or a depth-two RR cancellation directly on $J^{(2)}$. We do not pursue this direction here.

4. Richardson–Romberg PR Weight Bounds

4.1. PR-Averaged Error Decomposition and the RR Weight

The starting point for all subsequent estimates is the **explicit** representation of $\sqrt{n}(\bar{\theta}_n^{(\text{RR}, \alpha)} - \theta^*)$ as a noise-weighted sum with a deterministic kernel. This subsection derives that representation step by step.

Depth-one expansion. Unfolding the error recursion of Chapter 1 gives, for every $k \geq 1$,

$$\theta_k^{(\alpha)} - \theta^* = -\alpha \sum_{l=1}^k \Gamma_{l+1:k}^{(\alpha)} \epsilon(Z_l) + \Gamma_{1:k}^{(\alpha)} (\theta_0 - \theta^*), \quad \Gamma_{l+1:k}^{(\alpha)} := \prod_{j=l+1}^k (I - \alpha A(Z_j)).$$

Replacing each random product $\Gamma_{l+1:k}^{(\alpha)}$ by its deterministic counterpart $B_\alpha^{k-l} := (I - \alpha \bar{A})^{k-l}$ and absorbing the difference into a higher-order remainder yields the **depth-one decomposition** (Samsonov et al., 2025, Proposition 9):

$$\theta_k^{(\alpha)} - \theta^* = -\alpha \sum_{l=1}^k B_\alpha^{k-l} \epsilon(Z_l) + B_\alpha^k (\theta_0 - \theta^*) + R_k^{(\alpha)},$$

where $R_k^{(\alpha)} := J_k^{(1, \alpha)} + H_k^{(1, \alpha)}$ is the first misadjustment remainder. Its leading component $J_k^{(1, \alpha)}$ has a stationary bias of order α , so it is not treated as an $\alpha^{3/2}$ term. The Berry–Esseen proof below controls this remainder by refining it to depth two, where $J^{(2)} + H^{(2)}$ carries the $\alpha^{3/2}$ moment scale. The first sum is the **depth-zero** term and is the only piece that carries the limiting Gaussian.

PR averaging produces $Q_l^{(\alpha)}$. Recall the PR average $\bar{\theta}_n^{(\alpha)} = (n - n_0)^{-1} \sum_{k=n_0}^{n-1} \theta_k^{(\alpha)}$. For notational clarity, and for the theorem proved in this chapter, we set $n_0 = 0$ from this point on. This is not only a cosmetic convention: a burned-in average has a different deterministic weight,

$$Q_{l, n_0}^{(\alpha)} = \frac{n}{n - n_0} \alpha \sum_{k=\max(n_0, l)}^{n-1} B_\alpha^{k-l},$$

when the statistic is normalized as $-n^{-1/2} \sum_l Q_{l, n_0}^{(\alpha)} \epsilon(Z_l)$. Thus the burned-in non-stationary theorem requires redoing the weight, Poisson, and variance-comparison arguments with $Q_{l, n_0}^{(\alpha)}$. The present chapter proves the full-average result and keeps the deterministic initial-condition transient explicit in the final assembly. Subtracting θ^* , substituting the depth-one decomposition above, and **exchanging the order of summation** in the depth-zero piece,

$$\sum_{k=0}^{n-1} \sum_{l=1}^k B_\alpha^{k-l} \epsilon(Z_l) = \sum_{l=1}^{n-1} \left(\sum_{k=l}^{n-1} B_\alpha^{k-l} \right) \epsilon(Z_l),$$

isolates each noise sample $\epsilon(Z_l)$ together with the deterministic **PR weight**

$$Q_l^{(\alpha)} := \alpha \sum_{k=l}^{n-1} B_\alpha^{k-l} = \alpha \sum_{j=0}^{n-l-1} B_\alpha^j.$$

Operationally, $Q_l^{(\alpha)}$ collects the cumulative contribution of $\epsilon(Z_l)$ to all averaged iterates $\theta_l, \theta_{l+1}, \dots, \theta_{n-1}$. Multiplying the resulting identity by $\sqrt{n}$ gives the PR-averaged decomposition

$$\sqrt{n} (\bar{\theta}_n^{(\alpha)} - \theta^*) = W^{(\alpha)} + D^{(\alpha)},$$

where the **leading martingale-like sum** is

$$W^{(\alpha)} := -\frac{1}{\sqrt{n}} \sum_{l=1}^{n-1} Q_l^{(\alpha)} \epsilon(Z_l),$$

and the remainder $D^{(\alpha)}$ packs the deterministic transient $D_{\text{tr}}^{(\alpha)}$ and the higher-order stochastic part $D_R^{(\alpha)}$:

$$D^{(\alpha)} := D_{\text{tr}}^{(\alpha)} + D_R^{(\alpha)}, \quad D_{\text{tr}}^{(\alpha)} := \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} B_{\alpha}^k (\theta_0 - \theta^*), \quad D_R^{(\alpha)} := \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} R_k^{(\alpha)}.$$

The first sum is the deterministic initial-condition transient. Under the full-average convention it is only $O((\sqrt{n}\alpha)^{-1})$ in general, so it must either be retained explicitly or removed by a centered initialization $\theta_0 = \theta^*$. The second is the source of the leading non-Gaussian correction in the Berry–Esseen rate; its RR-combination $2D_R^{(\alpha)} - D_R^{(2\alpha)}$ is exactly the **misadjustment** $D_1^{\text{mis, RR}}$ diagnosed in Chapters 2 and 3 and controlled below through the Levin depth-two transfer.

RR combination produces $\mathcal{Q}_l^{\text{RR}}$. The Richardson–Romberg iterate $\bar{\theta}_n^{(\text{RR}, \alpha)} := 2\bar{\theta}_n^{(\alpha)} - \bar{\theta}_n^{(2\alpha)}$ inherits the PR decomposition **by linearity**: applying the previous display at step sizes α and 2α separately and combining,

$$\sqrt{n} (\bar{\theta}_n^{(\text{RR}, \alpha)} - \theta^*) = W^{\text{RR}} + D^{\text{RR}},$$

$$W^{\text{RR}} := 2W^{(\alpha)} - W^{(2\alpha)} = -\frac{1}{\sqrt{n}} \sum_{l=1}^{n-1} \mathcal{Q}_l^{\text{RR}} \epsilon(Z_l),$$

where the **RR weight** is

$$\mathcal{Q}_l^{\text{RR}} := 2\mathcal{Q}_l^{(\alpha)} - \mathcal{Q}_l^{(2\alpha)}.$$

Crucially, both PR averages share the **same** noise realization $\{Z_k\}$, so the bracket $\mathcal{Q}_l^{\text{RR}}$ is a single deterministic matrix kernel: all the stochastic content of W^{RR} now lives in $\epsilon(Z_l)$ alone.

Why $\mathcal{Q}_l^{\text{RR}}$ is the right object. Two questions about W^{RR} drive the rest of the chapter, and both answer themselves in terms of $\mathcal{Q}_l^{\text{RR}}$:

1. **Variance comparison.** The target CLT covariance of W^{RR} is $\Sigma_{\infty} = \bar{A}^{-1} \Sigma_{\epsilon}^{(M)} \bar{A}^{-\top}$ (the Markov-chain CLT covariance), which is exactly what one obtains if every $\mathcal{Q}_l^{\text{RR}}$ is replaced by the asymptotic weight $\bar{A}^{-1}$. The finite- n deviation is therefore controlled by $\sum_l \|\mathcal{Q}_l^{\text{RR}} - \bar{A}^{-1}\|^2$ — see Section 4.5.
2. **Poisson-equation / Abel-summation remainder.** The standard Berry–Esseen route for Markov-chain noise solves the Poisson equation $\hat{\epsilon} - Q\hat{\epsilon} = \epsilon$, replaces $\epsilon(Z_l)$ by $\hat{\epsilon}(Z_l) - \hat{\epsilon}(Z_{l+1})$ (up to a martingale increment), and Abel-sums against the weight sequence $\mathcal{Q}_l^{\text{RR}}$. The resulting remainder has norm bounded by the **total variation** $\sum_l \|\mathcal{Q}_{l+1}^{\text{RR}} - \mathcal{Q}_l^{\text{RR}}\|$.

Both quantities are estimated in Sections 4.3–4.4 below. The closed-form identities of Section 4.2 reduce them to elementary operator-norm calculations on $B_{\alpha}^m - B_{2\alpha}^m$.

4.2. Closed-Form Identities

Throughout this chapter write

$$B_{\alpha} := I - \alpha \bar{A}, \quad B_{2\alpha} := I - 2\alpha \bar{A}, \quad k := n - l.$$

We assume $\alpha, 2\alpha \in (0, \alpha_{\infty}]$, so that the Lyapunov contraction $\|B_{\alpha}^m\|_Q^2 \leq (1 - \alpha a)^m$ and $\|B_{2\alpha}^m\|_Q^2 \leq (1 - 2\alpha a)^m$ hold (and we use freely $1 - 2\alpha a \leq 1 - \alpha a$).

The geometric-series identity $\sum_{j=0}^{m-1} B_{\alpha}^j = (\alpha \bar{A})^{-1} (I - B_{\alpha}^m)$ converts into the closed form

$$\mathcal{Q}_l^{(\alpha)} = \alpha \sum_{j=0}^{n-l-1} B_{\alpha}^j = \alpha (\alpha \bar{A})^{-1} (I - B_{\alpha}^{n-l}) = \bar{A}^{-1} (I - B_{\alpha}^{n-l}).$$

This makes the convergence $Q_l^{(\alpha)} \rightarrow \bar{A}^{-1}$ as $k = n - l \rightarrow \infty$ explicit: the only obstruction is the geometric Lyapunov tail B_α^k . Two immediate consequences are

$$Q_l^{(\alpha)} - \bar{A}^{-1} = -\bar{A}^{-1} B_\alpha^{n-l}, \quad Q_{l+1}^{(\alpha)} - Q_l^{(\alpha)} = -\alpha B_\alpha^{n-l-1}.$$

The first is the **asymptotic-weight error** and decays geometrically in k . The second is the **discrete derivative** used in Abel summation; note the explicit factor α in front.

Applying both identities at α and 2α and combining yields the basic **RR identities**:

$$Q_l^{\text{RR}} - \bar{A}^{-1} = -\bar{A}^{-1} (2B_\alpha^k - B_{2\alpha}^k),$$

$$Q_{l+1}^{\text{RR}} - Q_l^{\text{RR}} = -2\alpha (B_\alpha^{k-1} - B_{2\alpha}^{k-1}).$$

These are the exact expressions bounded in the next subsection. Note the structural asymmetry between them:

- The asymptotic-weight error $2B_\alpha^k - B_{2\alpha}^k$ is at the **single-trajectory** rate $(1 - \alpha a)^{k/2}$ — the triangle inequality $|2X - Y| \leq 2|X| + |Y|$ does not see the RR coupling, and indeed the leading $-\bar{A}^{-1}$ in Q_l^{RR} is the same as in the single-step weight $Q_l^{(\alpha)}$.
- The discrete derivative $B_\alpha^{k-1} - B_{2\alpha}^{k-1}$, however, is a **true** difference of contractions evaluated at the same exponent. The elementary identity $X^m - Y^m = (X - Y) \sum_{i=1}^m X^{i-1} Y^{m-i}$ with $X = B_\alpha$, $Y = B_{2\alpha}$, $X - Y = \alpha \bar{A}$ extracts an extra factor of α , gaining one full power of α over the single-step case $Q_{l+1}^{(\alpha)} - Q_l^{(\alpha)} = -\alpha B_\alpha^{k-1}$.

This local α -gain in the discrete derivative is the **only** manifestation of Richardson–Romberg cancellation visible at the level of the PR weights.

4.3. Pointwise Bounds for the RR Weights

Lemma 7. Let $\alpha, 2\alpha \in (0, \alpha_\infty]$, set $C_Q := \kappa_Q^{1/2} \|\bar{A}^{-1}\|$ and $\tilde{C}_A := \kappa_Q \|\bar{A}\|$, and write $k = n - l$.

(i) For every $1 \leq l \leq n - 1$,

$$\|Q_l^{\text{RR}} - \bar{A}^{-1}\| \leq 3C_Q (1 - \alpha a)^{k/2}.$$

(ii) For every $1 \leq l \leq n - 2$,

$$\|Q_{l+1}^{\text{RR}} - Q_l^{\text{RR}}\| \leq 2\tilde{C}_A \alpha^2 (k - 1) (1 - \alpha a)^{(k-2)/2}.$$

Proof of (i). Take norms in the basic RR identity above. Using the equivalence $\|\cdot\| \leq \kappa_Q^{1/2} \|\cdot\|_Q$ and the Lyapunov contraction at both step sizes,

$$\|2B_\alpha^k - B_{2\alpha}^k\|_Q \leq 2\|B_\alpha^k\|_Q + \|B_{2\alpha}^k\|_Q \leq 2(1 - \alpha a)^{k/2} + (1 - 2\alpha a)^{k/2} \leq 3(1 - \alpha a)^{k/2}.$$

Submultiplicativity in $\|\cdot\|_Q$ followed by norm equivalence then gives

$$\|Q_l^{\text{RR}} - \bar{A}^{-1}\| \leq \kappa_Q^{1/2} \|\bar{A}^{-1}\| \cdot \|2B_\alpha^k - B_{2\alpha}^k\|_Q \leq 3C_Q (1 - \alpha a)^{k/2}.$$

Proof of (ii). Apply the elementary identity $X^m - Y^m = (X - Y) \sum_{i=1}^m X^{i-1} Y^{m-i}$ with $X = B_\alpha$, $Y = B_{2\alpha}$, $X - Y = \alpha \bar{A}$, and $m = k - 1$:

$$B_\alpha^{k-1} - B_{2\alpha}^{k-1} = \alpha \bar{A} \sum_{i=1}^{k-1} B_\alpha^{i-1} B_{2\alpha}^{k-1-i}.$$

Each summand obeys, by submultiplicativity in $\|\cdot\|_Q$ and Lyapunov contraction,

$$\begin{aligned}\|B_\alpha^{i-1} B_{2\alpha}^{k-1-i}\| &\leq \kappa_Q^{1/2} (1-\alpha a)^{(i-1)/2} (1-2\alpha a)^{(k-1-i)/2} \\ &\leq \kappa_Q^{1/2} (1-\alpha a)^{(k-2)/2},\end{aligned}$$

where in the last step we used $1-2\alpha a \leq 1-\alpha a$. Summing over $i \in \{1, \dots, k-1\}$ and absorbing constants,

$$\|B_\alpha^{k-1} - B_{2\alpha}^{k-1}\| \leq \alpha \kappa_Q \|\bar{A}\| (k-1)(1-\alpha a)^{(k-2)/2}.$$

Inserting this bound into the discrete-difference identity finishes the proof. $\square$

4.4. Summed Bounds and Comparison with the Single-Step Case

Corollary 1. Under the assumptions of the previous lemma, uniformly in $n \geq 2$,

$$\sum_{l=1}^{n-1} \|Q_l^{\text{RR}} - \bar{A}^{-1}\|^2 \leq \frac{C_1}{\alpha a}, \quad \sum_{l=1}^{n-2} \|Q_{l+1}^{\text{RR}} - Q_l^{\text{RR}}\| \leq \frac{C_2}{a^2},$$

with $C_1 = 9C_Q^2$ and $C_2 = 32\tilde{C}_A$.

Proof. For the first sum apply (i) and the geometric series $\sum_{k \geq 1} (1-\alpha a)^k \leq \frac{1}{\alpha a}$. For the second, apply (ii) and

$$\sum_{k \geq 2} (k-1)(1-\alpha a)^{(k-2)/2} = \sum_{m \geq 0} (m+1)(1-\alpha a)^{m/2} \leq \frac{1}{(1-\sqrt{1-\alpha a})^2} \leq \frac{4}{(\alpha a)^2},$$

where the last step uses $1-\sqrt{1-\alpha a} \geq \alpha a/2$ for $\alpha a \leq 1/2$. Multiplying by $2\tilde{C}_A \alpha^2$ gives the claim, up to the stated universal constant. $\square$

Remark 2. Comparison with the single-step case. The same identities for $Q_l^{(\alpha)}$ give $\|Q_l^{(\alpha)} - \bar{A}^{-1}\| \leq C_Q(1-\alpha a)^{k/2}$ and $\|Q_{l+1}^{(\alpha)} - Q_l^{(\alpha)}\| \leq \alpha \kappa_Q^{1/2} (1-\alpha a)^{(k-1)/2}$. Part (i) of the lemma above is therefore the same order as the single-step bound: the leading $-\bar{A}^{-1}$ term of Q_{n-1}^{RR} is exactly $-\bar{A}^{-1}$ (since $Q_{n-1}^{\text{RR}} = 0$ identically), so no factor of α is gained on the boundary. The Richardson–Romberg cancellation is visible in the successive difference of (ii): for fixed lag k the discrete derivative gains an extra factor of α . This local gain comes with the factor $k-1$, so after summing over the geometric tail it should not be read as a global extra factor of α in the Abel term. What is needed for the Poisson-equation remainder is the uniform total-variation bound in the corollary. On the variance side the corollary gives $\sum \|Q_l^{\text{RR}} - \bar{A}^{-1}\|^2 = O(\frac{1}{\alpha a})$, which yields the finite-horizon comparison error $O(\frac{1}{n\alpha a})$.

4.5. Variance Comparison

After the Poisson decomposition the $l=1$ noise sample is absorbed into the Poisson boundary remainder, while the martingale increments are indexed by $l \in \{2, \dots, n-1\}$. The deterministic variance proxy used in the martingale Berry–Esseen step must therefore use the same index set:

$$\Sigma_n^{\text{RR}} := \frac{1}{n} \sum_{l=2}^{n-1} Q_l^{\text{RR}} \Sigma_\epsilon^{(\text{M})} (Q_l^{\text{RR}})^\top, \quad \sigma_n^{2, \text{RR}}(u) := u^\top \Sigma_n^{\text{RR}} u,$$

with $\Sigma_\epsilon^{(\text{M})}$ the symmetric long-run noise covariance of the Markov chain,

$$\Sigma_\epsilon^{(\text{M})} = \mathbb{E}_\pi [\epsilon(Z_0) \epsilon(Z_0)^\top] + \sum_{j \geq 1} \left(\mathbb{E}_\pi [\epsilon(Z_0) \epsilon(Z_j)^\top] + \mathbb{E}_\pi [\epsilon(Z_j) \epsilon(Z_0)^\top] \right).$$

The asymptotic covariance is

$$\Sigma_\infty = \bar{A}^{-1} \Sigma_\epsilon^{(\text{M})} \bar{A}^{-\top}, \quad \sigma^2(u) = u^\top \Sigma_\infty u.$$

The bounds of the previous section give a quantitative comparison between Σ_n^{RR} and Σ_∞ .

Lemma 8. Let $\alpha, 2\alpha \in (0, \alpha_\infty]$, set $\Sigma := \Sigma_\epsilon^{(M)}$, and assume $\|\Sigma\| < \infty$. Then

$$\|\Sigma_n^{\text{RR}} - \Sigma_\infty\| \leq \frac{C_3}{n\alpha a},$$

with $C_3 = 12C_Q \|\bar{A}^{-1}\| \|\Sigma\| + 9C_Q^2 \|\Sigma\| + 2\|\Sigma_\infty\|$. Consequently, for every $u \in \mathbb{R}^d$,

$$|\sigma_n^{\text{RR}}(u) - \sigma^2(u)| \leq \frac{C_3 \|u\|^2}{n\alpha a}.$$

At the working scale $\alpha = c n^{-1/2}$ this is $O(n^{-1/2})$.

Proof. Write $\Delta_l := Q_l^{\text{RR}} - \bar{A}^{-1}$ and expand

$$Q_l^{\text{RR}} \Sigma (Q_l^{\text{RR}})^\top - \Sigma_\infty = \underbrace{\Delta_l \Sigma \bar{A}^{-\top} + \bar{A}^{-1} \Sigma \Delta_l^\top}_{R_{1,l}} + \underbrace{\Delta_l \Sigma \Delta_l^\top}_{R_{2,l}}.$$

Submultiplicativity of the operator norm yields the pointwise bounds

$$\|R_{1,l}\| \leq 2\|\bar{A}^{-1}\| \|\Sigma\| \|\Delta_l\|, \quad \|R_{2,l}\| \leq \|\Sigma\| \|\Delta_l\|^2.$$

Summing the linear part with the bound from part (i) of the previous lemma and the geometric series $\sum_{k \geq 1} (1 - \alpha a)^{k/2} \leq \frac{1}{1 - \sqrt{1 - \alpha a}} \leq \frac{2}{\alpha a}$ (valid for $\alpha a \leq 1/2$),

$$\sum_{l=1}^{n-1} \|\Delta_l\| \leq 3C_Q \sum_{k=1}^{n-1} (1 - \alpha a)^{k/2} \leq \frac{6C_Q}{\alpha a}.$$

The quadratic part is bounded directly by the previous corollary,

$$\sum_{l=1}^{n-1} \|\Delta_l\|^2 \leq \frac{9C_Q^2}{\alpha a}.$$

Since the sum starts at $l = 2$, replacing every weight by $\bar{A}^{-1}$ gives $((n-2)/n)\Sigma_\infty$, not Σ_∞ . Hence there is an additional deterministic finite-sum boundary term $2\Sigma_\infty/n$. Combining,

$$\begin{aligned} \|\Sigma_n^{\text{RR}} - \Sigma_\infty\| &\leq \frac{2\|\Sigma_\infty\|}{n} + \frac{1}{n} \sum_{l=2}^{n-1} (\|R_{1,l}\| + \|R_{2,l}\|) \\ &\leq \frac{1}{n} \left(2\|\bar{A}^{-1}\| \|\Sigma\| \cdot \frac{6C_Q}{\alpha a} + \|\Sigma\| \cdot \frac{9C_Q^2}{\alpha a} + 2\|\Sigma_\infty\|/(\alpha a) \right) = \frac{C_3}{n\alpha a}, \end{aligned}$$

where we used $\alpha a \leq 1$. This proves the operator-norm bound. The scalar bound on $|\sigma_n^{\text{RR}}(u) - \sigma^2(u)| = |u^\top (\Sigma_n^{\text{RR}} - \Sigma_\infty) u|$ follows from the Cauchy–Schwarz inequality. $\square$

Remark 3. The constant C_3 is independent of α and n ; only a enters through the Lyapunov contraction. The bound is tight in the geometric scale: the dominant contribution comes from the near-boundary region $l \approx n$, where $\|\Delta_l\|$ is bounded below by a positive constant (see the boundary identity $Q_{n-1}^{\text{RR}} = 0$). The same calculation for the single-step PR weights yields $\|\Sigma_n^{\text{PR}} - \Sigma_\infty\| \leq \frac{C}{n\alpha a}$, so Richardson–Romberg does not improve the variance comparison rate, but it also does not degrade it: the leading-order variance is preserved.

4.6. Poisson Martingale Approximation

The variance comparison of the previous section identifies the limiting covariance of W^{RR} , but it does not by itself produce a martingale. This subsection converts W^{RR} into a martingale plus a quantitatively small remainder via the Poisson equation for the Markov chain $\{Z_l\}$. The deterministic kernel bounds of Sections 4.3–4.4 enter once and for all in the boundary/Abel control of the remainder.

Poisson kernel. Let Q denote the one-step Markov transition kernel of $(Z_k)_{k \geq 1}$, acting on bounded measurable functions $f : Z \rightarrow \mathbb{R}^d$ by

$$(Qf)(z) = \mathbb{E}[f(Z_{k+1}) \mid Z_k = z].$$

Under UGE 1 with mixing time t_{mix} , the geometric Dobrushin bound $\|Q^k \epsilon\|_\infty \leq 2 \|\epsilon\|_\infty (1/4)^{\lfloor k/t_{\text{mix}} \rfloor}$ (valid for centered ϵ , $\pi(\epsilon) = 0$) makes the Poisson series

$$\hat{\epsilon} := \sum_{k=0}^{\infty} Q^k \epsilon$$

absolutely convergent in sup-norm, with

$$\|\hat{\epsilon}\|_\infty \leq 2 \|\epsilon\|_\infty \sum_{k=0}^{\infty} (1/4)^{\lfloor k/t_{\text{mix}} \rfloor} \leq 3 t_{\text{mix}} \|\epsilon\|_\infty, \quad \|Q\hat{\epsilon}\|_\infty \leq \|\hat{\epsilon}\|_\infty,$$

the second inequality because Q is a Markov kernel and contracts the sup-norm. By construction $\hat{\epsilon}$ solves the **Poisson equation**

$$\hat{\epsilon} - Q\hat{\epsilon} = \epsilon.$$

Conditional centering. Let $\mathcal{F}_l := \sigma(Z_1, \dots, Z_l)$. Substituting into the noise sample and adding and subtracting the conditional mean,

$$\epsilon(Z_l) = \underbrace{[\hat{\epsilon}(Z_l) - Q\hat{\epsilon}(Z_{l-1})]}_{\text{martingale increment}} + \underbrace{[Q\hat{\epsilon}(Z_{l-1}) - Q\hat{\epsilon}(Z_l)]}_{\text{telescope}}, \quad l \geq 2,$$

where the first bracket is centered conditionally on $\mathcal{F}_{l-1}$ since the Markov property gives $\mathbb{E}[\hat{\epsilon}(Z_l) \mid \mathcal{F}_{l-1}] = Q\hat{\epsilon}(Z_{l-1})$. The $l = 1$ term has no past state to condition on; we treat it directly via as $\epsilon(Z_1) = \hat{\epsilon}(Z_1) - Q\hat{\epsilon}(Z_1)$. The decomposition therefore separates W^{RR} into a martingale piece and a deterministic-coefficient telescope; Abel summation against the kernel sequence $\{Q_l^{\text{RR}}\}$ converts the telescope into boundary terms plus a sum against the discrete derivative $Q_{l+1}^{\text{RR}} - Q_l^{\text{RR}}$, which is exactly the object bounded in the Corollary of Section 4.4.

Lemma 9. Assume **UGE 1** and $\pi(\epsilon) = 0$. Set

$$\Delta M_l^{\text{RR}} := Q_l^{\text{RR}} (\hat{\epsilon}(Z_l) - Q\hat{\epsilon}(Z_{l-1})), \quad 2 \leq l \leq n-1,$$

and let $M_n^{\text{RR}} := \sum_{l=2}^{n-1} \Delta M_l^{\text{RR}}$. Then $\{\Delta M_l^{\text{RR}}\}_{l=2}^{n-1}$ is a sequence of $\mathcal{F}_l$ -martingale differences, and

$$W^{\text{RR}} = -\frac{1}{\sqrt{n}} M_n^{\text{RR}} + D_{2,n}^{\text{RR}},$$

with the **Poisson boundary/Abel remainder**

$$D_{2,n}^{\text{RR}} := -\frac{1}{\sqrt{n}} \left[Q_1^{\text{RR}} \hat{\epsilon}(Z_1) + \sum_{l=1}^{n-2} (Q_{l+1}^{\text{RR}} - Q_l^{\text{RR}}) Q\hat{\epsilon}(Z_l) \right].$$

(The right boundary $Q_{n-1}^{\text{RR}} Q\hat{\epsilon}(Z_{n-1})$ vanishes identically because $Q_{n-1}^{\text{RR}} = 0$.) Moreover, with $C_Q := \|\bar{A}^{-1}\| + 3C_Q$ a uniform bound on $\|Q_l^{\text{RR}}\|$, and C_2 the constant from the Corollary of Section 4.4,

$$\|D_{2,n}^{\text{RR}}\|_\infty \leq \frac{3 t_{\text{mix}} \|\epsilon\|_\infty}{\sqrt{n}} \left(C_Q + \frac{C_2}{a^2} \right).$$

Consequently, for every $p \geq 1$,

$$\| D_{2,n}^{\text{RR}} \|_{L_p} \leq \frac{C t_{\text{mix}} \| \epsilon \|_{\infty}}{a^2 \sqrt{n}},$$

with a constant C depending only on $\| \bar{A}^{-1} \|$, κ_Q , and $\| \bar{A} \|$.

Proof. The increments ΔM_l^{RR} are $\mathcal{F}_l$ -martingale differences because Q_l^{RR} is deterministic and the Markov property gives $\mathbb{E}[\hat{\epsilon}(Z_l) \mid \mathcal{F}_{l-1}] = Q\hat{\epsilon}(Z_{l-1})$.

For the decomposition, substitute into the definition :

$$W^{\text{RR}} = -\frac{1}{\sqrt{n}} \sum_{l=1}^{n-1} Q_l^{\text{RR}} (\hat{\epsilon}(Z_l) - Q\hat{\epsilon}(Z_l)).$$

Peel off $l = 1$ and apply the conditional-centering identity above to the remaining $l \in \{2, \dots, n-1\}$:

$$W^{\text{RR}} = -\frac{1}{\sqrt{n}} Q_1^{\text{RR}} \hat{\epsilon}(Z_1) + \frac{1}{\sqrt{n}} Q_1^{\text{RR}} Q\hat{\epsilon}(Z_1) - \frac{1}{\sqrt{n}} M_n^{\text{RR}} - \frac{1}{\sqrt{n}} \sum_{l=2}^{n-1} Q_l^{\text{RR}} (Q\hat{\epsilon}(Z_{l-1}) - Q\hat{\epsilon}(Z_l)).$$

Set $g_l := Q\hat{\epsilon}(Z_l)$ and Abel-sum the last sum:

$$\sum_{l=2}^{n-1} Q_l^{\text{RR}} (g_{l-1} - g_l) = Q_2^{\text{RR}} g_1 - Q_{n-1}^{\text{RR}} g_{n-1} + \sum_{l=2}^{n-2} (Q_{l+1}^{\text{RR}} - Q_l^{\text{RR}}) g_l.$$

Adding the contribution $Q_1^{\text{RR}} g_1$ from the peeled $l = 1$ term combines with $-Q_2^{\text{RR}} g_1$ to yield $-(Q_2^{\text{RR}} - Q_1^{\text{RR}}) g_1$, which folds the boundary $l = 1$ summand back into the discrete-derivative sum:

$$\begin{aligned} & -Q_1^{\text{RR}} \hat{\epsilon}(Z_1) + Q_1^{\text{RR}} g_1 - Q_2^{\text{RR}} g_1 + Q_{n-1}^{\text{RR}} g_{n-1} - \sum_{l=2}^{n-2} (Q_{l+1}^{\text{RR}} - Q_l^{\text{RR}}) g_l \\ & = -Q_1^{\text{RR}} \hat{\epsilon}(Z_1) - \sum_{l=1}^{n-2} (Q_{l+1}^{\text{RR}} - Q_l^{\text{RR}}) g_l + Q_{n-1}^{\text{RR}} g_{n-1}. \end{aligned}$$

The rightmost boundary $Q_{n-1}^{\text{RR}} g_{n-1}$ vanishes because $Q_{n-1}^{\text{RR}} = 2\alpha I - 2\alpha I = 0$ identically (from $Q_{n-1}^{(\alpha)} = \alpha I$ and $Q_{n-1}^{(2\alpha)} = 2\alpha I$). Multiplying by $1/\sqrt{n}$ and absorbing the sign into $D_{2,n}^{\text{RR}}$ produces exactly the stated formula.

For the sup-norm bound, use $\| g_l \|_{\infty} \leq \| \hat{\epsilon} \|_{\infty} \leq 3t_{\text{mix}} \| \epsilon \|_{\infty}$, the uniform bound $\| Q_1^{\text{RR}} \| \leq C_Q$ on the left boundary, and the summed-total-variation bound $\sum_{l=1}^{n-2} \| Q_{l+1}^{\text{RR}} - Q_l^{\text{RR}} \| \leq C_2/a^2$ on the Abel sum:

$$\| \sqrt{n} D_{2,n}^{\text{RR}} \| \leq 3t_{\text{mix}} \| \epsilon \|_{\infty} (C_Q + C_2/a^2).$$

The L_p bound is immediate from the deterministic sup-norm bound. $\square$

Remark 4. Two structural features. First, the bound is **deterministic**: no martingale concentration or moment inequality is invoked, only pointwise bounds on Q_l^{RR} , on its discrete derivative, and on $\hat{\epsilon}$. The dependence on p promised in the working plan ($\text{poly}(p, t_{\text{mix}})$) is therefore absorbed into the constant; the effective scaling is $\| D_{2,n}^{\text{RR}} \|_{L_p} \leq C t_{\text{mix}} / (a^2 \sqrt{n})$. Second, the $1/a^2$ factor is **exactly** the cost of the discrete derivative summed Abel-style: it is the same C_2/a^2 that appears in the second part of the Corollary of Section 4.4. No further $1/\alpha$ blow-up enters, because the kernel-difference bound of Section 4.3 already carries the cancelling $\alpha^2(k-1)$ prefactor.

Boundary asymmetry. The right-end boundary vanishes because $Q_{n-1}^{\text{RR}} = 0$ — a Richardson–Romberg cancellation that does not occur for the single-step PR weight $Q_{n-1}^{(\alpha)} = \alpha I$, which contributes a (small) $\alpha t_{\text{mix}}/\sqrt{n}$ residue. RR therefore **removes** one of the two Poisson-boundary terms outright.

Comparison with martingale scale. The leading martingale piece $M_n^{\text{RR}}/\sqrt{n}$ has L_2 -norm of order 1 (it is the natural $\sqrt{n}$ -scaled object whose CLT variance equals $\sigma_n^2, \text{RR}(u)$ at leading order, by the variance-

comparison lemma). The remainder $D_{2,n}^{\text{RR}}$ is therefore $O(n^{-1/2})$ relative to it, well below the target Berry–Esseen scale $n^{-1/4}$.

4.7. Predictable Quadratic Variation Concentration

The Berry–Esseen step for the martingale M_n^{RR} requires a quantitative control of its predictable quadratic variation $\langle M^{\text{RR}} \rangle_n$ around its deterministic asymptotic counterpart $n\sigma_n^{2,\text{RR}}(u)$. This subsection produces such a control by applying the Markov concentration result of Section 2.2 to a suitable centered quadratic functional of the chain. The result is the RR analogue of Lemmas 22–23 of Samsonov et al. (2025).

Predictable quadratic variation. By the Poisson Martingale Approximation lemma above, the increments $\Delta M_l^{\text{RR}} = \mathcal{Q}_l^{\text{RR}}(\hat{\epsilon}(Z_l) - Q\hat{\epsilon}(Z_{l-1}))$ are $\mathcal{F}_l$ -martingale differences. The Markov property $\mathbb{E}[\hat{\epsilon}(Z_l) \mid \mathcal{F}_{l-1}] = Q\hat{\epsilon}(Z_{l-1})$ gives, by direct computation,

$$\begin{aligned} \mathbb{E}[\Delta M_l^{\text{RR}} (\Delta M_l^{\text{RR}})^\top \mid \mathcal{F}_{l-1}] &= \mathcal{Q}_l^{\text{RR}} \mathbb{E}[\hat{\epsilon}(Z_l) \hat{\epsilon}(Z_l)^\top \mid \mathcal{F}_{l-1}] (\mathcal{Q}_l^{\text{RR}})^\top \\ &\quad - \mathcal{Q}_l^{\text{RR}} Q\hat{\epsilon}(Z_{l-1}) (Q\hat{\epsilon})(Z_{l-1})^\top (\mathcal{Q}_l^{\text{RR}})^\top \\ &= \mathcal{Q}_l^{\text{RR}} \bar{\epsilon}(Z_{l-1}) (\mathcal{Q}_l^{\text{RR}})^\top, \end{aligned}$$

where the cross-terms in the expansion of $(\hat{\epsilon}(Z_l) - Q\hat{\epsilon}(Z_{l-1}))(\hat{\epsilon}(Z_l) - Q\hat{\epsilon}(Z_{l-1}))^\top$ cancel by the Markov property and we have set

$$\bar{\epsilon}(z) := Q(\hat{\epsilon}\hat{\epsilon}^\top)(z) - (Q\hat{\epsilon})(z)(Q\hat{\epsilon})(z)^\top.$$

Summing over $l \in \{2, \dots, n-1\}$ and tracking that $\langle M^{\text{RR}} \rangle_n := \sum_{l=2}^{n-1} \mathbb{E}[\Delta M_l^{\text{RR}} (\Delta M_l^{\text{RR}})^\top \mid \mathcal{F}_{l-1}]$,

$$\langle M^{\text{RR}} \rangle_n = \sum_{l=2}^{n-1} \mathcal{Q}_l^{\text{RR}} \bar{\epsilon}(Z_{l-1}) (\mathcal{Q}_l^{\text{RR}})^\top.$$

The function $\bar{\epsilon}$ has stationary mean equal to the long-run noise covariance,

$$\pi(\bar{\epsilon}) = \Sigma_\epsilon^{(\text{M})} =: \Sigma,$$

which is a standard identity for the Poisson solution of a Markov chain (Samsonov et al. 2025, Eq. (10); see also Douc–Moulines–Priouret–Soulie 2018, Theorem 21.2.5).

Sup-norm bound on $\bar{\epsilon}$. Since $\|\hat{\epsilon}\|_\infty \leq 3t_{\text{mix}}\|\epsilon\|_\infty$ and Q is a Markov kernel (so Qf inherits the sup-norm of f),

$$\begin{aligned} \|\bar{\epsilon}\|_\infty &\leq \|Q(\hat{\epsilon}\hat{\epsilon}^\top)\|_\infty + \|(Q\hat{\epsilon})(Q\hat{\epsilon})^\top\|_\infty \\ &\leq 2\|\hat{\epsilon}\|_\infty^2 \leq 18t_{\text{mix}}^2\|\epsilon\|_\infty^2. \end{aligned}$$

Lemma 10. Assume **UGE 1** and $\pi(\epsilon) = 0$, with $\|\epsilon\|_\infty < \infty$. Let C_Q be the uniform bound on $\|\mathcal{Q}_l^{\text{RR}}\|$ from the previous lemma. There exists a universal constant $C_4 > 0$ such that, for every $u \in \mathbb{R}^d$, every $p \geq 2$, every initial distribution ξ , and every $n \geq 2$,

$$\mathbb{E}_\xi^{1/p}[\|u^\top \langle M^{\text{RR}} \rangle_n u - n\sigma_n^{2,\text{RR}}(u)\|^p] \leq C_4 C_Q^2 \|u\|^2 \|\epsilon\|_\infty^2 t_{\text{mix}}^{5/2} \sqrt{pn}.$$

Proof. Write $h_{l(z)} := u^\top \mathcal{Q}_l^{\text{RR}} \bar{\epsilon}(z) (\mathcal{Q}_l^{\text{RR}})^\top u$ and $g_{l(z)} := h_{l(z)} - \pi(h_l)$. Submultiplicativity of the operator norm and give

$$|h_{l(z)}| \leq C_Q^2 \|u\|^2 \|\bar{\epsilon}\|_\infty \leq 18C_Q^2 \|u\|^2 t_{\text{mix}}^2 \|\epsilon\|_\infty^2, \quad \|g_l\|_\infty \leq 2|h_{l(z)}| \leq 36C_Q^2 \|u\|^2 t_{\text{mix}}^2 \|\epsilon\|_\infty^2.$$

By and the variance definition $n\sigma_n^{2,\text{RR}}(u) = \sum_{l=2}^{n-1} \pi(h_l)$,

$$u^\top \langle M^{\text{RR}} \rangle_n u - n \sigma_n^{2, \text{RR}}(u) = \sum_{l=2}^{n-1} g_{l(Z_{l-1})}.$$

For the centered sum, set $\tilde{g}_i := g_{i+1}$ for $i \in \{1, \dots, n-2\}$. By construction $\pi(\tilde{g}_i) = 0$ and $\|\tilde{g}_i\|_\infty \leq c := 36 C_Q^2 \|u\|^2 t_{\text{mix}}^2 \|\epsilon\|_\infty^2$. The Markov concentration lemma of Section 2.2 (Levin et al. 2025, Lemma 11) applied to $\sum_{i=1}^{n-2} \tilde{g}_{i(Z_i)}$ yields, for every $t \geq 0$,

$$\mathbb{P}_\xi \left(\left| \sum_{i=1}^{n-2} \tilde{g}_{i(Z_i)} \right| \geq t \right) \leq 2 \exp \left(-\frac{t^2}{2 u_n^2} \right), \quad u_n^2 \leq 64 t_{\text{mix}} (n-2) c^2.$$

The sub-Gaussian-to-moment lemma of Section 2.2 then gives, for $p \geq 2$,

$$\mathbb{E}_\xi^{1/p} \left[\left| \sum_{l=2}^{n-1} g_{l(Z_{l-1})} \right|^p \right] \leq 2^{1/p} \sqrt{p} u_n \leq 2\sqrt{p} u_n \leq 16\sqrt{2} \sqrt{p(n-2)} c \sqrt{t_{\text{mix}}}.$$

Substituting c and bounding $\sqrt{n-2} \leq \sqrt{n}$,

$$\mathbb{E}_\xi^{1/p} \left[\left| \sum_{l=2}^{n-1} g_{l(Z_{l-1})} \right|^p \right] \leq 16\sqrt{2} \cdot 36 C_Q^2 \|u\|^2 \|\epsilon\|_\infty^2 t_{\text{mix}}^{5/2} \sqrt{p n} = 576\sqrt{2} C_Q^2 \|u\|^2 \|\epsilon\|_\infty^2 t_{\text{mix}}^{5/2} \sqrt{p n}.$$

Rounding the resulting prefactor to $C_4 := 850$ gives $\square$.

Corollary 2. Under the assumptions of the previous lemma, for every $u \in \mathbb{R}^d$, every $p \geq 2$, and every $n \geq 2$,

$$\mathbb{E}_\xi^{1/p} [|u^\top \langle M^{\text{RR}} \rangle_n u - n \sigma^2(u)|^p] \leq C_4 C_Q^2 \|u\|^2 \|\epsilon\|_\infty^2 t_{\text{mix}}^{5/2} \sqrt{p n} + \frac{C_3 \|u\|^2}{\alpha a},$$

with C_3 the variance-comparison constant of Section 4.5.

Proof. The triangle inequality and the variance-comparison lemma give

$$|u^\top \langle M^{\text{RR}} \rangle_n u - n \sigma^2(u)| \leq |u^\top \langle M^{\text{RR}} \rangle_n u - n \sigma_n^{2, \text{RR}}(u)| + n |\sigma_n^{2, \text{RR}}(u) - \sigma^2(u)|.$$

The first piece is bounded in L_p by ; the second is the deterministic bound $n C_3 \|u\|^2 / (n \alpha a) = C_3 \|u\|^2 / (\alpha a)$. $\square$

Remark 5. Working scale and Berry–Esseen consequence. The bound $|u^\top \langle M^{\text{RR}} \rangle_n u - n \sigma_n^{2, \text{RR}}(u)|_{L_p} \leq C t_{\text{mix}}^{5/2} \sqrt{p n}$ is exactly what feeds into the martingale Berry–Esseen of Bolthausen–Fan type (cf. Lemma 21 / Proposition 13 of Samsonov et al. 2025): the conditional-variance term in the Bolthausen–Fan inequality is $\sqrt{p} s_n^{-2p/(2p+1)} \mathbb{E}^{1/(2p+1)} [\sum \sigma_l^2 - s_n^2]^p$, which with $s_n^2 = n \sigma_n^{2, \text{RR}}(u) \asymp n$ (uniformly in α at the working scale) and the exponent $\sqrt{p n} \rightarrow n^{p/(2p+1)}$ gives the standard $n^{-1/4}$ rate after taking $p = \lceil \log n \rceil$.

RR vs single-step structure. The bound is structurally identical to the single-step PR case (Samsonov et al. 2025, Lemma 23): the only RR-specific input is the uniform sup bound $\|Q_l^{\text{RR}}\| \leq C_Q$. No extra $1/a$ or $1/\alpha$ factor enters, because the Markov concentration is applied to a **bounded** function and the weights Q_l^{RR} enter only through their sup-norm. Variance comparison (Section 4.5) is the only place where $1/(\alpha a)$ appears, and at the working scale $\alpha = c n^{-1/2}$ this contributes $O(\sqrt{n})$ — the same order as the leading concentration term.

Why we centred against $\sigma_n^{2, \text{RR}}(u)$. The natural normalisation in the Bolthausen–Fan martingale BE step is $s_n^2 = \sum_l \sigma_l^2$, which is precisely $n \sigma_n^{2, \text{RR}}(u)$ for the increments ΔM_l^{RR} . Centering against $n \sigma^2(u)$ (corollary above) is used in the corollary that converts the BE statement from the $\sigma_n^{\text{RR}}(u)$ normalisation to the asymptotic $\sigma(u)$ normalisation.

4.8. Martingale Berry–Esseen Step

The Poisson decomposition (Section 4.6) writes $W^{\text{RR}} = -n^{-1/2} M_n^{\text{RR}} + D_{2,n}^{\text{RR}}$, with the remainder $D_{2,n}^{\text{RR}}$ controlled in sup-norm at the order $n^{-1/2}$. The quadratic variation concentration (Section 4.7) bounds $|u^\top \langle M^{\text{RR}} \rangle_n u - n \sigma_n^2(u)|$ in L_p by $C \sqrt{p n}$. This subsection assembles those two inputs into a Berry–Esseen rate for the **scalar** martingale $u^\top M_n^{\text{RR}}$, the leading contribution to the final RR Berry–Esseen bound.

Bounded increments. Fix $u \in \mathbb{R}^d$ and set $X_l := u^\top \Delta M_l^{\text{RR}}$ for $2 \leq l \leq n-1$. The X_l are $\mathcal{F}_l$ -martingale differences (Section 4.6), and bounding the three factors $\|u\|$, $\|Q_l^{\text{RR}}\| \leq C_Q$, and $\|\hat{\epsilon}(Z_l) - Q\hat{\epsilon}(Z_{l-1})\| \leq 2\|\hat{\epsilon}\|_\infty \leq 6t_{\text{mix}}\|\epsilon\|_\infty$ yields the deterministic bound

$$|X_l| \leq \varkappa(u), \quad \varkappa(u) := 6t_{\text{mix}} C_Q \|\epsilon\|_\infty \|u\|.$$

The conditional variances $\sigma_l^2 := \mathbb{E}[X_l^2 \mid \mathcal{F}_{l-1}]$ sum to the predictable quadratic variation along u ,

$$V_n^2 := \sum_{l=2}^{n-1} \sigma_l^2 = u^\top \langle M^{\text{RR}} \rangle_n u.$$

Variance lower bound. Set $s_n^2 := n \sigma_n^2(u)$. The variance comparison lemma (Section 4.5) and assumption $\sigma^2(u) > 0$ give $|\sigma_n^2(u) - \sigma^2(u)| \leq C_3 \|u\|^2 / (n\alpha a)$, so whenever

$$n\alpha a \geq \frac{2C_3 \|u\|^2}{\sigma^2(u)}$$

one has $\sigma_n^2(u) \geq \sigma^2(u)/2$, i.e. $s_n^2 \geq n\sigma^2(u)/2$. At the working scale $\alpha = c n^{-1/2}$, condition is satisfied for $n \geq (2C_3 \|u\|^2 / (c a \sigma^2(u)))^2$. The trivial upper bound $\sigma_n^2(u) \leq C_Q^2 \|\Sigma_\epsilon^{(\text{M})}\| \|u\|^2$ gives $s_n^2 \leq K^2(u) n$ with $K(u) := C_Q \|\Sigma_\epsilon^{(\text{M})}\|^{1/2} \|u\|$.

Bolthausen–Fan inequality. We apply Lemma 21 of Samsonov et al. (2025), which combines Bolthausen (1982) with Fan (2019). For any sequence of $\mathcal{F}_l$ -martingale differences $\{X_l\}$ with $|X_l| \leq \varkappa$, any $p \geq 1$,

$$\begin{aligned} d_K(S_n/s_n, \mathcal{N}(0, 1)) &\leq L_{B(\varkappa)} \frac{(2n+1) \log(2n+1)}{s_n^3} \\ &\quad + C_1 \sqrt{p} s_n^{-2p/(2p+1)} (\mathbb{E} |V_n^2 - s_n^2|^p)^{1/(2p+1)} \\ &\quad + C_2 s_n^{-2p/(2p+1)} p \varkappa^{2p/(2p+1)}, \end{aligned}$$

with $S_n := \sum_l X_l$, $L_{B(\varkappa)}$ the finite bounded-increment constant appearing in Samsonov et al.’s statement, and C_1, C_2 universal.

Theorem 1. Assume **UGE 1**, $\pi(\epsilon) = 0$, $\|\epsilon\|_\infty < \infty$, $\sigma^2(u) > 0$, and $\alpha, 2\alpha \in (0, \alpha_\infty]$. There exist constants $C_{K,1}(u), C_{K,2}(u) > 0$ depending only on $\|u\|$, $\sigma(u)$, C_Q , t_{mix} , $\|\epsilon\|_\infty$, $\|\Sigma_\epsilon^{(\text{M})}\|$, and the constants $L_{B(\varkappa(u))}, C_1, C_2$ of , such that for every $n \geq 3$ satisfying ,

$$d_K\left(\frac{u^\top M_n^{\text{RR}}}{\sqrt{n} \sigma_n^{\text{RR}}(u)}, \mathcal{N}(0, 1)\right) \leq \frac{C_{K,1}(u) \log^{3/4} n}{n^{1/4}} + \frac{C_{K,2}(u) \log n}{\sqrt{n}}.$$

Proof. Apply with $S_n = u^\top M_n^{\text{RR}}$, the increment bound $\varkappa = \varkappa(u)$ from , $s_n^2 = n \sigma_n^2(u)$, and $p = \lceil \log n \rceil$ (so $\log n \leq p \leq \log n + 1$). Bound the three terms in turn, using $s_n^2 \in [n \sigma^2(u)/2, K^2(u) n]$ from the variance lower and upper bounds.

Term I (classical Bolthausen). From $s_n^3 \geq (n \sigma^2(u)/2)^{3/2}$ and $(2n+1) \log(2n+1) \leq 3 n \log n$ for $n \geq 3$,

$$L_{B(\varkappa(u))} \frac{(2n+1) \log(2n+1)}{s_n^3} \leq \frac{6\sqrt{2} L_{B(\varkappa(u))} \log n}{\sigma^3(u) \sqrt{n}} =: \frac{C^{(\text{I})}(u) \log n}{\sqrt{n}}.$$

Term III (Lindeberg). Write $a_p := 2p/(2p+1) = 1 - 1/(2p+1)$. Using $s_n^{-a_p} = s_n^{-1} s_n^{1/(2p+1)}$ and $s_n \leq K(u) \sqrt{n}$,

$$s_n^{1/(2p+1)} \leq \max(1, K(u))^{1/(2p+1)} n^{1/(2(2p+1))}.$$

For $p \geq \log n$ one has $n^{1/(2(2p+1))} \leq n^{1/(4 \log n)} = e^{1/4}$, similarly $\max(1, K(u))^{1/(2p+1)} \leq e^{1/2}$ once $n \geq \max(1, K(u))$, which is absorbed into $C^{(\text{III})}(u)$. Bounding $\varkappa(u)^{a_p} \leq \max(1, \varkappa(u))$ and $s_n^{-1} \leq \sqrt{2}/(\sigma(u)\sqrt{n})$,

$$C_2 s_n^{-a_p} p \varkappa(u)^{a_p} \leq \sqrt{2} e^{3/4} \frac{C_2 \max(1, \varkappa(u))}{\sigma(u)} \frac{p}{\sqrt{n}} \leq \frac{C^{(\text{III})}(u) \log n}{\sqrt{n}},$$

with $C^{(\text{III})}(u)$ depending on $\|u\|, \sigma(u), C_Q, \|\Sigma_\epsilon^{(\text{M})}\|, t_{\text{mix}}, \|\epsilon\|_\infty$ and on C_2 .

Term II (conditional-variance concentration). By Lemma 4.7 (equation), for every $p \geq 2$,

$$(\mathbb{E} | V_n^2 - s_n^2 |^p)^{1/p} \leq B(u) \sqrt{p n}, \quad B(u) := C_4 C_Q^2 \|u\|^2 \|\epsilon\|_\infty^2 t_{\text{mix}}^{5/2},$$

hence $(\mathbb{E} | V_n^2 - s_n^2 |^p)^{1/(2p+1)} \leq B(u)^{p/(2p+1)} (p n)^{p/(2(2p+1))}$. Combining with $s_n^{-1} \leq \sqrt{2}/(\sigma^2(u)n)$ and $s_n^{1/(2p+1)} \leq \max(1, K(u))^{1/(2p+1)} n^{1/(2(2p+1))}$,

$$\begin{aligned} & C_1 \sqrt{p} s_n^{-a_p} (\mathbb{E} | V_n^2 - s_n^2 |^p)^{1/(2p+1)} \\ & \leq C_1 \sqrt{p} \sqrt{2/(\sigma^2(u)n)} \max(1, K(u))^{1/(2p+1)} n^{1/(2(2p+1))} \\ & \quad \times B(u)^{p/(2p+1)} p^{p/(2(2p+1))} n^{p/(2(2p+1))} \\ & = C_1 \sqrt{2/\sigma^2(u)} \max(1, K(u))^{1/(2p+1)} B(u)^{p/(2p+1)} p^{(3p+1)/(2(2p+1))} n^{-p/(2(2p+1))}. \end{aligned}$$

At $p = \lceil \log n \rceil \geq \log n$, three elementary bounds hold:

$$\begin{aligned} p^{(3p+1)/(2(2p+1))} & \leq p^{3/4} \leq 2^{3/4} (\log n)^{3/4}, \\ n^{-p/(2(2p+1))} & = n^{-1/4+1/(4(2p+1))} \leq e^{1/8} n^{-1/4}, \end{aligned}$$

$$B(u)^{p/(2p+1)} \leq \max(1, \sqrt{B(u)}) e^{1/2},$$

where the first uses $(3p+1)/(2(2p+1)) \leq 3/4$ (equivalent to $4(3p+1) \leq 6(2p+1)$, true), the second uses $\log(n^{1/(4(2p+1))}) = \log(n)/(4(2p+1)) \leq 1/8$ for $p \geq \log n$, and the third uses $|p/(2p+1) - 1/2| \leq 1/(2(2p+1))$ together with $\log B(u)/(2p+1) \leq 1$ for $n \geq \max(1, B(u))$ (absorbed into $C^{(\text{II})}(u)$). Likewise $\max(1, K(u))^{1/(2p+1)} \leq e^{1/2}$ for $n \geq K(u)$. Substituting,

$$\begin{aligned} \text{Term II} & \leq C_1 \sqrt{2/\sigma^2(u)} e^{1/2} \max(1, \sqrt{B(u)}) e^{1/2} 2^{3/4} (\log n)^{3/4} e^{1/8} n^{-1/4} \\ & \leq 2^{3/4} e^{9/8} C_1 \sqrt{2 \max(1, B(u))/\sigma^2(u)} \frac{\log^{3/4} n}{n^{1/4}} =: \frac{C^{(\text{II})}(u) \log^{3/4} n}{n^{1/4}}. \end{aligned}$$

Adding , , and setting $C_{K,1}(u) := C^{(\text{II})}(u)$, $C_{K,2}(u) := C^{(\text{I})}(u) + C^{(\text{III})}(u)$ proves . $\square$

Corollary 3. Under the hypotheses of the previous theorem,

$$d_K \left(\frac{u^\top M_n^{\text{RR}}}{\sqrt{n} \sigma(u)}, \mathcal{N}(0, 1) \right) \leq \frac{C_{K,1}(u) \log^{3/4} n}{n^{1/4}} + \frac{C_{K,2}(u) \log n}{\sqrt{n}} + \frac{C_3 \|u\|^2}{n \alpha \sigma^2(u)}.$$

At $\alpha = c n^{-1/2}$ the last term is $O(n^{-1/2})$, hence absorbed into $C_{K,2}(u) \log n / \sqrt{n}$ up to a constant.

Proof. Set $r := \sigma_n^{\text{RR}}(u)/\sigma(u)$ and $W := u^\top M_n^{\text{RR}}/(\sqrt{n} \sigma_n^{\text{RR}}(u))$. Then the statistic normalised by $\sigma(u)$ is Wr . Under , $r \geq 1/\sqrt{2}$, while the trivial upper bound on $\sigma_n^{\text{RR}}(u)$ gives $r \leq r_{\max(u)} < \infty$. On this compact interval the standard normal cdf satisfies

$$\sup_x |\Phi(x/r) - \Phi(x)| \leq C_\Phi |r - 1|, \quad C_\Phi := \sqrt{2}/\sqrt{\pi e}.$$

Therefore

$$d_{K(Wr, \mathcal{N}(0,1))} \leq d_{K(W, \mathcal{N}(0,1))} + C_\Phi |r - 1|.$$

The variance comparison of Section 4.5 gives

$$|r - 1| = \frac{|\sigma_n^{2, \text{RR}}(u) - \sigma^2(u)|}{\sigma(u) (\sigma_n^{\text{RR}}(u) + \sigma(u))} \leq \frac{C_3 \|u\|^2}{n \alpha a \sigma^2(u)}.$$

Adding this perturbation to Theorem proves the stated bound after absorbing C_Φ into the constants. $\square$

Remark 6. Working scale and rate. At $\alpha = c n^{-1/2}$ the leading rate $\log^{3/4}(n) n^{-1/4}$ matches the single-step PR Berry–Esseen of Samsonov et al. (2025, Proposition 13). Richardson–Romberg neither improves nor degrades the martingale Berry–Esseen rate; the cancellation acts on the **misadjustment** remainder, not on the leading martingale.

Constants. The dominant $C_{K,1}(u)$ scales as $\sqrt{B(u)/\sigma^2(u)}$ with $B(u)$ the conditional-variance concentration constant of : explicitly $C_{K,1}(u) \asymp C_Q \|u\|^2 t_{\text{mix}}^{5/4} \|\epsilon\|_\infty / \sigma(u)$. The $\sigma(u)^{-1}$ blow-up at degenerate directions is genuine and reflects the assumption $\sigma^2(u) > 0$.

Scope. This theorem bounds the Kolmogorov distance for the martingale $u^\top M_n^{\text{RR}}$ alone, not for the full PR-averaged RR iterate $\sqrt{n} u^\top (\bar{\theta}_n^{(\text{RR}, \alpha)} - \theta^*)$. The full Berry–Esseen bound additionally requires L_p -control of $D_{2,n}^{\text{RR}}$ (Section 4.6, already done), the transient $D_{\text{tr}}^{\text{RR}}$, and the misadjustment $R_n^{\text{mis, RR}}$ via the Levin depth-two transfer of the next subsection, assembled by the smoothing inequality of Samsonov et al. (2025, Proposition 12).

4.9. Misadjustment via Levin Depth-Two

The Berry–Esseen assembly so far controls the leading martingale piece (Section 4.8) and the Poisson boundary remainder $D_{2,n}^{\text{RR}}$ (Section 4.6). The remaining non-martingale contribution to $\sqrt{n} u^\top (\bar{\theta}_n^{(\text{RR}, \alpha)} - \theta^*)$ is the **PR-averaged misadjustment**

$$R_n^{\text{mis, RR}} := \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} (2R_k^{(\alpha)} - R_k^{(2\alpha)}),$$

where $R_k^{(\alpha)} := J_k^{(1, \alpha)} + H_k^{(1, \alpha)}$ is the depth-one remainder of . Chapter 3 showed that the kernel-difference identity bounds $R_n^{\text{mis, RR}}$ at order $O(\sqrt{n} \alpha) = O(1)$ at the working scale $\alpha \asymp n^{-1/2}$ — too crude for a Berry–Esseen rate $n^{-1/4}$. The depth-two route below transfers four statements of Levin et al. (2025) into the present notation and recovers the target $n^{-1/4} \text{polylog}(n)$ rate.

Depth-two refinement. The deterministic-product expansion of Levin et al. extends the depth-one decomposition by one more level. For $\ell \geq 1$ define

$$J_n^{(\ell, \alpha)} := (I - \alpha \bar{A}) J_{n-1}^{(\ell, \alpha)} - \alpha \tilde{A}(Z_n) J_{n-1}^{(\ell-1, \alpha)},$$

$$H_n^{(\ell, \alpha)} := (I - \alpha A(Z_n)) H_{n-1}^{(\ell, \alpha)} - \alpha \tilde{A}(Z_n) J_{n-1}^{(\ell, \alpha)},$$

with $J_0^{(\ell, \alpha)} = H_0^{(\ell, \alpha)} = 0$ and the $\ell = 0$ processes as in `last_iterate.typ`. Substituting $A(Z_n) = \bar{A} + \tilde{A}(Z_n)$ and grouping terms gives, by induction on n , the recursive identity

$$H_n^{(\ell, \alpha)} = J_n^{(\ell+1, \alpha)} + H_n^{(\ell+1, \alpha)}, \quad \ell \geq 0.$$

Applying with $\ell = 1$ refines $R_k^{(\alpha)}$ to

$$R_k^{(\alpha)} = J_k^{(1,\alpha)} + J_k^{(2,\alpha)} + H_k^{(2,\alpha)},$$

which splits the misadjustment into three structurally different pieces:

$$R_n^{\text{mis, RR}} = T_n^{(1)} + T_n^{(2)} + T_n^{(H)},$$

$$T_n^{(j)} := \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} \left(2J_k^{(j,\alpha)} - J_k^{(j,2\alpha)} \right), \quad j \in \{1, 2\},$$

with $T_n^{(H)}$ defined identically with $H^{(2)}$ in place of $J^{(j)}$. Each piece is bounded separately below.

Cited inputs. The depth-two route imports four statements of Levin et al. (2025), specialized to the constant step-size LSA setting. They are stated here for reference; their proofs are in the cited paper.

Lemma 11. (Levin Proposition 2 — stationary bias of $J^{(1)}$.) Under stationarity for the augmented chain $(Z_{t+1}, J_t^{(0,\alpha)}, J_t^{(1,\alpha)})$,

$$\mathbb{E}_\pi [J_\infty^{(1,\alpha)}] = \alpha \Delta + R(\alpha), \quad \|R(\alpha)\| \leq 12 \|\bar{A}^{-1}\| C_A^2 t_{\text{mix}}^2 \alpha^2 \|\epsilon\|_\infty,$$

with $\Delta := \bar{A}^{-1} \sum_{k \geq 1} \mathbb{E}_\pi [\left(Q^k \tilde{A} \right) (Z_0) \epsilon(Z_0)]$.

Lemma 12. (Levin Corollary 6 — centered bilinear L_p bound.) Define $\bar{\psi}_\alpha(j, z) := \tilde{A}(z)j - \mathbb{E}_{\Pi_{J^{(0)}, \alpha}} [\tilde{A}(Z_1) J_0^{(0,\alpha)}]$. For every initial distribution, every $r \geq 1$, and every $p \geq 2$,

$$\left\| \sum_{t=0}^{r-1} \bar{\psi}_\alpha \left(J_t^{(0,\alpha)}, Z_{t+1} \right) \right\|_{L_p} \leq c_{W,1} p^{3/2} \sqrt{\alpha r} + c_{W,2} p^3 \alpha^{-1/2} \log^{1/p}(1/(\alpha a)),$$

with $c_{W,1}, c_{W,2}$ depending only on $C_A, \kappa_Q, t_{\text{mix}}, \|\epsilon\|_\infty$. The precise logarithmic factor is not rate-critical below; it is absorbed into $\text{polylog}(n)$ in the working-scale corollaries.

Lemma 13. (Levin Propositions 8 and 9 — high-order moment bounds.) For every $q \geq 2$ and every p satisfying $2 \leq p \leq q/2$, under the step-size restriction $\alpha \leq \alpha_*(q, t_{\text{mix}})$ of Levin et al. (2025),

$$\|J_n^{(2,\alpha)}\|_{L_p} \leq D_J t_{\text{mix}}^{5/2} p^{7/2} \alpha^{3/2} \log^{3/2}(1/(\alpha a)),$$

$$\|H_n^{(2,\alpha)}\|_{L_p} \leq D_H d^{1/q} t_{\text{mix}}^{5/2} p^{7/2} \alpha^{3/2} \log^{3/2}(1/(\alpha a)),$$

uniformly in n , with D_J, D_H depending only on $C_A, \kappa_Q, \|\bar{A}^{-1}\|, \|\epsilon\|_\infty$.

Stationary augmented-chain convention. The recursions above are displayed in finite-time notation with $J_0^{(\ell,\alpha)} = H_0^{(\ell,\alpha)} = 0$, while Levin Proposition 2 is a statement about the stationary augmented chain. The misadjustment estimates below use the stationary quantities as deterministic centers and should be read under the stationary augmented-chain convention. For a zero-start full average, the transfer from finite-start sums to stationary centered sums is not a single terminal ρ^n term: summing pointwise contractions over $k = 0, \dots, n-1$ produces a startup contribution of order

$$\frac{1}{\sqrt{n}} \sum_{k \geq 0} \rho_\alpha^k \asymp \frac{1}{\sqrt{n} \alpha a}$$

when $\rho_\alpha \approx \exp(-ca\alpha)$. This term is not part of the theorem proved here. A practical arbitrary-start theorem requires either a genuine burn-in average with the burned-in weights $Q_{l,n_0}^{(\alpha)}$ of Section 4.1 or a separate transfer lemma with the correct accumulated startup dependence.

Telescoping identity for $J^{(1)}$. Summing the recursion $J_k^{(1,\alpha)} = (I - \alpha \bar{A}) J_{k-1}^{(1,\alpha)} - \alpha \tilde{A}(Z_k) J_{k-1}^{(0,\alpha)}$ from $k = 1$ to n and rearranging, with $J_0^{(1,\alpha)} = 0$,

$$\bar{A} \sum_{k=0}^{n-1} J_k^{(1,\alpha)} = - \sum_{k=1}^n \tilde{A}(Z_k) J_{k-1}^{(0,\alpha)} - \frac{1}{\alpha} J_n^{(1,\alpha)}.$$

The stationary version of the same recursion yields $\mathbb{E}_\pi \left[\tilde{A}(Z_1) J_0^{(0,\alpha)} \right] = -\bar{A} \mathbb{E}_\pi \left[J_\infty^{(1,\alpha)} \right]$. Subtracting $n \mathbb{E}_\pi \left[J_\infty^{(1,\alpha)} \right]$ from both sides and applying $\bar{A}^{-1}$ gives the **centered telescoping identity**

$$\sum_{k=0}^{n-1} \left(J_k^{(1,\alpha)} - \mathbb{E}_\pi \left[J_\infty^{(1,\alpha)} \right] \right) = -\bar{A}^{-1} \sum_{k=1}^n \bar{\psi}_\alpha \left(J_{k-1}^{(0,\alpha)}, Z_k \right) - \frac{1}{\alpha} \bar{A}^{-1} J_n^{(1,\alpha)}.$$

This identity is the bridge between the (vector-valued) PR-average of $J^{(1)}$ and the centered bilinear sum bounded in Levin Corollary 6.

Lemma 14. Assume the stationary augmented-chain convention above, **UGE 1**, $\pi(\epsilon) = 0$, $\|\epsilon\|_\infty < \infty$, and $\alpha, 2\alpha \in (0, \alpha_\infty]$. Set

$$\Phi(p, \alpha) := p^{3/2} t_{\text{mix}}^{1/2} / a + p^{1/2} t_{\text{mix}}^{3/2} \sqrt{\alpha/a}.$$

There exists a constant $C_{\text{mis},1}$ depending on $\|\bar{A}\|, \|\bar{A}^{-1}\|, \kappa_Q, C_A, \|\epsilon\|_\infty, t_{\text{mix}}$, the universals $c_{W,1}, c_{W,2}$ of Levin Corollary 6, and the constant of the centered bound in `last_iterate.typ`, such that for every $p \geq 2$ and every $n \geq 2$,

$$\begin{aligned} \|T_n^{(1)}\|_{L_p} &\leq C_{\text{mis},1} \sqrt{n} \alpha^2 + C_{\text{mis},1} p^{3/2} \sqrt{\alpha} \\ &\quad + C_{\text{mis},1} p^3 (\alpha n)^{-1/2} \log^{1/p}(1/(\alpha a)) + C_{\text{mis},1} \Phi(p, \alpha) n^{-1/2}. \end{aligned}$$

Proof. Decompose $T_n^{(1)} = T_n^{(1,b)} + T_n^{(1,c)}$ via the bias-fluctuation split

$$T_n^{(1,b)} := \sqrt{n} \left(2 \mathbb{E}_\pi \left[J_\infty^{(1,\alpha)} \right] - \mathbb{E}_\pi \left[J_\infty^{(1,2\alpha)} \right] \right),$$

$$T_n^{(1,c)} := \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} \sum_{w \in \{\alpha, 2\alpha\}} c_w \left(J_k^{(1,w)} - \mathbb{E}_\pi \left[J_\infty^{(1,w)} \right] \right), \quad c_\alpha = 2, c_{2\alpha} = -1.$$

The bias is deterministic; the centered piece carries all the L_p fluctuation.

Bias. By Levin Proposition 2 applied at $w \in \{\alpha, 2\alpha\}$, $\mathbb{E}_\pi \left[J_\infty^{(1,w)} \right] = w \Delta + R(w)$ with $\|R(w)\| \leq 12 \|\bar{A}^{-1}\| C_A^2 t_{\text{mix}}^2 w^2 \|\epsilon\|_\infty$. The leading $w \Delta$ cancels in the RR combination, $2\alpha \Delta - 2\alpha \Delta = 0$, leaving

$$\|2 \mathbb{E}_\pi \left[J_\infty^{(1,\alpha)} \right] - \mathbb{E}_\pi \left[J_\infty^{(1,2\alpha)} \right]\| \leq 2 \|R(\alpha)\| + \|R(2\alpha)\| \leq 72 \|\bar{A}^{-1}\| C_A^2 t_{\text{mix}}^2 \alpha^2 \|\epsilon\|_\infty.$$

Multiplying by $\sqrt{n}$ produces the first term of the bound.

Centered. Apply at $w \in \{\alpha, 2\alpha\}$, take L_p norms, and combine via the triangle inequality:

$$\begin{aligned} \|T_n^{(1,c)}\|_{L_p} &\leq \frac{1}{\sqrt{n}} \sum_{w \in \{\alpha, 2\alpha\}} |c_w| \left(\|\bar{A}^{-1}\| \cdot \left\| \sum_{k=1}^n \bar{\psi}_w \left(J_{k-1}^{(0,w)}, Z_k \right) \right\|_{L_p} \right. \\ &\quad \left. + \frac{1}{w} \|\bar{A}^{-1}\| \cdot \|J_n^{(1,w)}\|_{L_p} \right). \end{aligned}$$

For the bilinear sum apply Levin Corollary 6 with $r = n$,

$$\left\| \sum_{k=1}^n \bar{\psi}_w \right\|_{L_p} \leq c_{W,1} p^{3/2} \sqrt{wn} + c_{W,2} p^3 w^{-1/2} \log^{1/p}(1/(wa)),$$

and divide by $\sqrt{n}$, using $w \in [\alpha, 2\alpha]$ to upper-bound by the α -form,

$$\frac{1}{\sqrt{n}} \left\| \sum \bar{\psi}_w \right\|_{L_p} \leq \sqrt{2} c_{W,1} p^{3/2} \sqrt{\alpha} + \sqrt{2} c_{W,2} p^3 (\alpha n)^{-1/2} \log^{1/p}(1/(\alpha a)).$$

For the boundary term, the centered last-iterate lemma in `last_iterate.typ`, applied to $T_n^{(1,w)} = (I - w\bar{A}) J_n^{(1,w)}$ with $\|(I - w\bar{A})^{-1}\| \leq 2$ for $w \leq \alpha_\infty$, gives

$$\|J_n^{(1,w)} - \mathbb{E}J_n^{(1,w)}\|_{L_p} \leq Cw \Phi(p, w),$$

and $\|\mathbb{E}J_n^{(1,w)}\| \leq w \|\Delta\| + \|R(w)\| \leq Cw$ by Levin Proposition 2; combining,

$$\|J_n^{(1,w)}\|_{L_p} \leq Cw(\Phi(p, w) + 1) \leq Cw \Phi(p, w) \quad \text{for } p \geq 2.$$

Therefore $w^{-1} \|J_n^{(1,w)}\|_{L_p} \leq C \Phi(p, w) \leq \sqrt{2} C \Phi(p, \alpha)$. Dividing by $\sqrt{n}$ produces the last term. Summing the three RR-combined pieces and absorbing universal factors into a single $C_{\text{mis},1}$ completes the proof. $\square$

Lemma 15. Under the assumptions of the previous lemma, for every $p \geq 2$ and every $q \geq 2$ satisfying $p \leq q/2$ and $\alpha \leq \alpha_*(q, t_{\text{mix}})$,

$$\|T_n^{(2)}\|_{L_p} + \|T_n^{(H)}\|_{L_p} \leq C_{\text{mis},2} (1 + d^{1/q}) p^{7/2} t_{\text{mix}}^{5/2} \sqrt{n} \alpha^{3/2} \log^{3/2}(1/(\alpha a)),$$

with $C_{\text{mis},2} := 6\sqrt{2}^3 (D_J + D_H)$.

Proof. Triangle inequality on the n summands of $T_n^{(2)}$, then Levin Propositions 8 and 9 applied uniformly in k and $w \in \{\alpha, 2\alpha\}$:

$$\begin{aligned} \|T_n^{(2)}\|_{L_p} &\leq \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} \sum_{w \in \{\alpha, 2\alpha\}} |c_w| \|J_k^{(2,w)}\|_{L_p} \\ &\leq 3\sqrt{n} \sup_{w \in \{\alpha, 2\alpha\}} \|J_k^{(2,w)}\|_{L_p} \\ &\leq 3D_J (2\alpha)^{3/2} t_{\text{mix}}^{5/2} p^{7/2} \sqrt{n} \log^{3/2}(1/(\alpha a)), \end{aligned}$$

where $|c_\alpha| + |c_{2\alpha}| = 3$ absorbs the RR combination and $(2\alpha)^{3/2}$ the worse bound at the larger step. Identical argument for $T_n^{(H)}$, with the additional $d^{1/q}$ factor of Levin Proposition 9. Adding the two bounds gives the lemma. $\square$

Theorem 2. Assume **UGE 1**, $\pi(\epsilon) = 0$, $\|\epsilon\|_\infty < \infty$, $\alpha, 2\alpha \in (0, \alpha_\infty]$, and the stationary augmented-chain convention above. There exists a constant C depending on the universal and problem constants of the previous two lemmas such that for every $p \geq 2$, every $q \geq 2$ satisfying $p \leq q/2$, every $\alpha \leq \alpha_*(q, t_{\text{mix}})$, and every $n \geq 2$,

$$\begin{aligned} \|R_n^{\text{mis, RR}}\|_{L_p} &\leq C\sqrt{n} \alpha^2 + C(1 + d^{1/q}) p^{7/2} t_{\text{mix}}^{5/2} \sqrt{n} \alpha^{3/2} \log^{3/2}(1/(\alpha a)) \\ &\quad + C p^{3/2} \sqrt{\alpha} + C p^3 (\alpha n)^{-1/2} \log^{1/p}(1/(\alpha a)) + C \Phi(p, \alpha) n^{-1/2}. \end{aligned}$$

Proof. Triangle inequality on $\|R_n^{\text{mis, RR}}\|_{L_p} \leq \|T_n^{(1)}\|_{L_p} + \|T_n^{(2)}\|_{L_p} + \|T_n^{(H)}\|_{L_p}$. Combine the centered $T^{(1)}$ bound and the raw $T^{(2)} + T^{(H)}$ bound. $\square$

Corollary 4. At the working scale $\alpha = c n^{-1/2}$, with $p = \lceil \log n \rceil$ and $q = \max(2p, \lceil \log(ed) \rceil, 2)$, and for n large enough that $\alpha \leq \alpha_*(q, t_{\text{mix}})$,

$$\|R_n^{\text{mis, RR}}\|_{L_p} \leq C \text{polylog}(n) n^{-1/4},$$

matching the leading martingale Berry–Esseen rate of Section 4.8.

Proof. At $\alpha = c n^{-1/2}$: $\sqrt{n} \alpha^2 = c^2 n^{-1/2}$, $\sqrt{n} \alpha^{3/2} = c^{3/2} n^{-1/4}$, $\sqrt{\alpha} = c^{1/2} n^{-1/4}$, $(\alpha n)^{-1/2} = c^{-1/2} n^{-1/4}$, and $\Phi(p, \alpha) n^{-1/2} = O(p^{3/2} n^{-1/2})$. With $p \asymp \log n$ and $q \geq \log(ed)$, the factor $d^{1/q}$ is bounded by a universal constant, and the dominant order in the misadjustment theorem is $\text{polylog}(n) n^{-1/4}$. $\square$

Remark 7. Finite-start transfer is not included. The misadjustment theorem is stated under the stationary augmented-chain convention introduced above. For a zero-start recursion or an arbitrary initial distribution ξ , the Wasserstein contraction of $(Z_t, J_t^{(0,\alpha)}, J_t^{(1,\alpha)})$ (Levin et al. 2025, Section 4 / Proposition 1) must be summed over the averaging window. For the full average $n_0 = 0$ this accumulated startup contribution is generally of order $1/(\sqrt{n}\alpha a)$ and is not negligible at $\alpha \asymp n^{-1/2}$. A logarithmic-burn-in theorem is therefore a separate extension and must be proved with the corresponding burned-in weights.

Why the depth-two route is needed. Chapter 3 proved that a kernel-difference identity bounds the centered RR-difference of S_n at order $O(\sqrt{n}\alpha) = O(1)$ at the working scale, which is too crude for a Berry–Esseen bound at rate $n^{-1/4}$. The depth-two refinement avoids this by separately controlling

1. the **bias** of $J^{(1)}$, where RR-cancellation drops the leading $\alpha \Delta$ to leave $O(\alpha^2)$ — Levin Proposition 2;
2. the **centered** part of $J^{(1)}$, where the recursion-telescoping identity reduces it to the centered bilinear bound of Levin Corollary 6;
3. the $J^{(2)}$ and $H^{(2)}$ pieces, controlled without RR-cancellation since the $\alpha^{3/2}$ moment bound of Levin Propositions 8 and 9 is already small enough.

Optional refinement. The depth-two analysis above does not exploit any RR-coupling between the α - and 2α -trajectories at the level of $J^{(2)}$ and $H^{(2)}$; the bound treats them as the sum of two independent $\alpha^{3/2}$ contributions. A genuine RR-cancellation at this level would replace $\sqrt{n}\alpha^{3/2}$ by $\sqrt{n}\alpha^2$ in the misadjustment theorem, making the misadjustment **strictly** subleading to the martingale Berry–Esseen. This is the open thread discussed at the end of Chapter 3.

4.10. Smoothing Assembly

The previous sections produced the four ingredients of the Berry–Esseen program for the PR-averaged Richardson–Romberg iterate:

1. the **Poisson decomposition** $W^{\text{RR}} = -n^{-1/2}M_n^{\text{RR}} + D_{2,n}^{\text{RR}}$ with deterministic sup-norm control of $D_{2,n}^{\text{RR}}$ at the order $t_{\text{mix}}/(a^2\sqrt{n})$ (Section 4.6, equation);
2. the **predictable quadratic variation concentration** of $u^\top \langle M^{\text{RR}} \rangle_n u$ around $n\sigma_n^{2,\text{RR}}(u)$ in L_p at the order $\sqrt{p n}$ (Lemma);
3. the **martingale Berry–Esseen** for $u^\top M_n^{\text{RR}}/(\sqrt{n}\sigma_n^{\text{RR}}(u))$ at rate $\log^{3/4}(n)n^{-1/4}$ (Theorem);
4. the **Levin depth-two misadjustment bound** on $R_n^{\text{mis,RR}}$ controlling the non-martingale residual at the same $\log^{c(n)}n^{-1/4}$ rate (Theorem).

The smoothing inequality of Bobkov–Götze (Samsonov et al. 2025, Proposition 12) assembles these four into a single Kolmogorov bound on $\sqrt{n}u^\top (\bar{\theta}_n^{(\text{RR},\alpha)} - \theta^*)$.

Composite remainder. Combining the depth-one identity in Section 4.1 with the Poisson decomposition of Section 4.6 gives

$$\sqrt{n}u^\top (\bar{\theta}_n^{(\text{RR},\alpha)} - \theta^*) = -\frac{u^\top M_n^{\text{RR}}}{\sqrt{n}} + u^\top \mathcal{R}_n^{\text{RR}},$$

$$\mathcal{R}_n^{\text{RR}} := D_{2,n}^{\text{RR}} + D_{\text{tr}}^{\text{RR}} + R_n^{\text{mis,RR}},$$

where $D_{2,n}^{\text{RR}}$ is the Poisson boundary/Abel remainder of Section 4.6, $D_{\text{tr}}^{\text{RR}} := 2D_{\text{tr}}^{(\alpha)} - D_{\text{tr}}^{(2\alpha)}$ is the deterministic transient of ‘s parent decomposition in Section 4.1, and $R_n^{\text{mis,RR}}$ is the Levin depth-two misadjustment defined in . Dividing by $\sqrt{n}\sigma_n^{\text{RR}}(u)$,

$$\frac{\sqrt{n}u^\top (\bar{\theta}_n^{(\text{RR},\alpha)} - \theta^*)}{\sigma_n^{\text{RR}}(u)} = X_n + Y_n,$$

$$X_n := -\frac{u^\top M_n^{\text{RR}}}{\sqrt{n} \sigma_n^{\text{RR}}(u)}, \quad Y_n := \frac{u^\top \mathcal{R}_n^{\text{RR}}}{\sigma_n^{\text{RR}}(u)}.$$

Theorem bounds $d_{K(X_n, \mathcal{N}(0,1))}$; the symmetry $-\mathcal{N}(0,1) \stackrel{d}{=} \mathcal{N}(0,1)$ of the standard normal makes the sign of the leading term irrelevant ($d_{K(X_n, \mathcal{N})} = d_{K(-X_n, \mathcal{N})}$).

Smoothing inequality. For random variables X, Y on the same probability space and any $t > 0$,

$$d_K(X + Y, \mathcal{N}(0,1)) \leq d_K(X, \mathcal{N}(0,1)) + \mathbb{P}(|Y| > t) + \frac{t}{\sqrt{2\pi}},$$

where the third term uses the $1/\sqrt{2\pi}$ Lipschitz constant of the standard normal cdf (Bobkov–Götze; see Samsonov et al. 2025, Proposition 12 for an LSA-tailored statement). Bounding $\mathbb{P}(|Y| > t) \leq \|Y\|_{L_p}^p / t^p$ via Markov’s inequality and choosing $t = e \|Y\|_{L_p}$ gives the **working form**

$$d_K(X + Y, \mathcal{N}(0,1)) \leq d_K(X, \mathcal{N}(0,1)) + \frac{e \|Y\|_{L_p}}{\sqrt{2\pi}} + e^{-p},$$

in which the trailing tail probability e^{-p} is absorbed into $O(1/n)$ as soon as $p \geq \log n$.

L_p -bound on the composite remainder. The three pieces of $\mathcal{R}_n^{\text{RR}}$ contribute additively. Consistently with the weight derivation of Section 4.1, this assembly is for the full average $n_0 = 0$. The deterministic transient is therefore not hidden inside a burn-in argument; it remains as an explicit term and disappears in the working-rate corollary under the centered initialization $\theta_0 = \theta^*$.

Lemma 16. Assume the standing hypotheses of Theorem . Then for every $u \in \mathbb{R}^d$, every $p \geq 2$, and every $q \geq 2$ satisfying the restrictions of Theorem ,

$$\begin{aligned} \|u^\top \mathcal{R}_n^{\text{RR}}\|_{L_p} &\leq \frac{C_{\text{D2}} \|u\|}{\sqrt{n}} + \frac{C_{\text{tr}} \|u\| \|\theta_0 - \theta^*\|}{\sqrt{n} \alpha a} \\ &\quad + \|u\| C_{\text{mis}} \left(\sqrt{n} \alpha^2 + (1 + d^{1/q}) p^{7/2} t_{\text{mix}}^{5/2} \sqrt{n} \alpha^{3/2} \log^{3/2}(1/(\alpha a)) \right. \\ &\quad \left. + p^{3/2} \sqrt{\alpha} + p^3 (\alpha n)^{-1/2} \log^{1/p}(1/(\alpha a)) + \Phi(p, \alpha) n^{-1/2} \right), \end{aligned}$$

with constants

$$C_{\text{D2}} := 3 t_{\text{mix}} \|\epsilon\|_\infty (C_Q + C_2/a^2), \quad C_{\text{tr}} := 6 \kappa_Q,$$

and C_{mis} the constant of Theorem .

Proof. Triangle inequality on $u^\top \mathcal{R}_n^{\text{RR}}$:

(a) The deterministic sup-norm bound for $D_{2,n}^{\text{RR}}$ yields $\|u^\top D_{2,n}^{\text{RR}}\|_{L_p} \leq \|u\| \|D_{2,n}^{\text{RR}}\|_\infty \leq C_{\text{D2}} \|u\| / \sqrt{n}$ for every $p \geq 1$.

(b) The deterministic transient $D_{\text{tr}}^{(\alpha)}$ of Section 4.1 is $D_{\text{tr}}^{(\alpha)} = n^{-1/2} \sum_{k=0}^{n-1} B_\alpha^k(\theta_0 - \theta^*)$. Submultiplicativity of $\|\cdot\|_Q$ and Lyapunov contraction give $\|\sum_{k=0}^{n-1} B_\alpha^k\|_Q \leq \sum_{k=0}^{n-1} (1 - \alpha a)^{k/2} \leq 2/(\alpha a)$, so by norm equivalence

$$\|D_{\text{tr}}^{(\alpha)}\| \leq \frac{2 \kappa_Q}{\sqrt{n} \alpha a} \|\theta_0 - \theta^*\|.$$

The RR combination $D_{\text{tr}}^{\text{RR}} = 2D_{\text{tr}}^{(\alpha)} - D_{\text{tr}}^{(2\alpha)}$ contributes a factor at most 3, giving $\|D_{\text{tr}}^{\text{RR}}\| \leq 6 \kappa_Q \|\theta_0 - \theta^*\| / (\sqrt{n} \alpha a)$, hence $\|u^\top D_{\text{tr}}^{\text{RR}}\| \leq C_{\text{tr}} \|u\| \|\theta_0 - \theta^*\| / (\sqrt{n} \alpha a)$.

(c) The misadjustment bound is Theorem , which gives the third group of summands directly. $\square$

Theorem 3. Under the standing hypotheses of Theorem and Theorem , with $p = \lceil \log n \rceil$ and $q = \max(2p, \lceil \log(e d) \rceil, 2)$, and with n large enough that holds and $\alpha \leq \alpha_*(q, t_{\text{mix}})$,

$$d_K \left(\frac{\sqrt{n} u^\top (\bar{\theta}_n^{(\text{RR}, \alpha)} - \theta^*)}{\sigma_n^{\text{RR}}(u)}, \mathcal{N}(0, 1) \right) \leq \frac{C_{K,1}(u) \log^{3/4} n}{n^{1/4}} + \frac{C_{K,2}(u) \log n}{\sqrt{n}} \\ + \frac{e \| u^\top \mathcal{R}_n^{\text{RR}} \|_{L_p}}{\sqrt{2\pi} \sigma_n^{\text{RR}}(u)} + \frac{e}{n},$$

with $\| u^\top \mathcal{R}_n^{\text{RR}} \|_{L_p}$ bounded by Lemma , and $C_{K,1}(u), C_{K,2}(u)$ the constants of Theorem .

Proof. Apply with $X = X_n$ and $Y = Y_n$ from . The first two terms come from Theorem applied to $-X_n$ (using $d_{K(X_n, \mathcal{N})} = d_{K(-X_n, \mathcal{N})}$ by symmetry of the standard normal). The remainder term is $e \| Y_n \|_{L_p} / \sqrt{2\pi} = e \| u^\top \mathcal{R}_n^{\text{RR}} \|_{L_p} / (\sqrt{2\pi} \sigma_n^{\text{RR}}(u))$. The trailing e^{-p} satisfies $e^{-p} \leq e n^{-1}$ for $p = \lceil \log n \rceil$. $\square$

Corollary 5. At the working scale $\alpha = c n^{-1/2}$ with $c > 0$ such that $\alpha, 2\alpha \in (0, \alpha_\infty]$, with $p = \lceil \log n \rceil$ and $q = \max(2p, \lceil \log(e d) \rceil, 2)$ satisfying $\alpha \leq \alpha_*(q, t_{\text{mix}})$, and under the centered initialization $\theta_0 = \theta^*$, the bound reduces to

$$d_K \left(\frac{\sqrt{n} u^\top (\bar{\theta}_n^{(\text{RR}, \alpha)} - \theta^*)}{\sigma_n^{\text{RR}}(u)}, \mathcal{N}(0, 1) \right) \leq \frac{C(u) \text{polylog}(n)}{n^{1/4}},$$

where $C(u)$ depends on $\| u \|$, $\sigma(u)$, C_Q , $\| \bar{A} \|$, $\| \bar{A}^{-1} \|$, κ_Q , C_A , $\| \epsilon \|_\infty$, $\| \Sigma_\epsilon^{(\text{M})} \|$, t_{mix} , a , α_∞ , c , and the universal constants of the smoothing, Bolthausen–Fan, and Levin Markov-concentration inequalities.

Proof. Substitute $\alpha = c n^{-1/2}$ in Lemma and bound each term separately: $C_{D2} \| u \| / \sqrt{n} = O(n^{-1/2})$; the deterministic transient is zero because $\theta_0 = \theta^*$; $\sqrt{n} \alpha^2 = c^2 n^{-1/2}$; $p^{7/2} \sqrt{n} \alpha^{3/2} \log^{3/2}(1/(\alpha a)) = O(\text{polylog}(n) n^{-1/4})$ at $p = \lceil \log n \rceil$; $p^{3/2} \sqrt{\alpha} = O(\log^{3/2}(n) n^{-1/4})$; $p^3 (\alpha n)^{-1/2} \log^{1/p}(1/(\alpha a)) = O(\log^3(n) n^{-1/4})$; $\Phi(p, \alpha) n^{-1/2} = O(\log^{3/2}(n) n^{-1/2})$. Adding and dividing by $\sigma_n^{\text{RR}}(u) \geq \sigma(u)/\sqrt{2}$ (from), the dominant order is $\text{polylog}(n) n^{-1/4}$. Combining with the leading $\log^{3/4}(n) n^{-1/4}$ Berry–Esseen of Theorem gives the claim. $\square$

Corollary 6. Under the hypotheses of Corollary , the same bound (with a different constant $C'(u)$) holds when the empirical normalisation $\sigma_n^{\text{RR}}(u)$ is replaced by the asymptotic $\sigma(u)$:

$$d_K \left(\frac{\sqrt{n} u^\top (\bar{\theta}_n^{(\text{RR}, \alpha)} - \theta^*)}{\sigma(u)}, \mathcal{N}(0, 1) \right) \leq \frac{C'(u) \text{polylog}(n)}{n^{1/4}}.$$

Proof. Set $r := \sigma_n^{\text{RR}}(u)/\sigma(u)$ and write $W := \sqrt{n} u^\top (\bar{\theta}_n^{(\text{RR}, \alpha)} - \theta^*)/\sigma_n^{\text{RR}}(u)$, so $Wr = \sqrt{n} u^\top (\bar{\theta}_n^{(\text{RR}, \alpha)} - \theta^*)/\sigma(u)$. Under , $r \in [1/\sqrt{2}, r_{\text{max}}]$ for some finite r_{max} (from the trivial upper bound $\sigma_n^{2, \text{RR}}(u) \leq C_Q^2 \| \Sigma \| \| u \|^2$). For r in this compact interval, the standard normal cdf satisfies

$$\sup_x |\Phi(x/r) - \Phi(x)| \leq C_\Phi |r - 1|, \quad C_\Phi := \sqrt{2}/\sqrt{\pi e},$$

where the constant arises from the maximum of $|x \varphi(x)|$ over $x \in \mathbb{R}$ being $1/\sqrt{2\pi e}$ (cf. the proof of Lemma 9 of Samsonov et al. 2025). Hence

$$d_K(Wr, \mathcal{N}(0, 1)) \leq d_K(W, \mathcal{N}(0, 1)) + C_\Phi |r - 1|.$$

The variance comparison of Section 4.5 gives

$$|r - 1| = \frac{|\sigma_n^{2, \text{RR}}(u) - \sigma^2(u)|}{\sigma(u) (\sigma_n^{\text{RR}}(u) + \sigma(u))} \leq \frac{C_3 \| u \|^2}{n \alpha a \sigma^2(u)} = O(n^{-1/2})$$

at the working scale, using $\sigma_n^{\text{RR}}(u) + \sigma(u) \geq \sigma(u)$. Adding this to Corollary proves the claim. $\square$

Remark 8. Full average vs. burn-in. Theorem is a full-average ($n_0 = 0$) statement. The working-rate corollary removes the deterministic initial-condition transient by assuming $\theta_0 = \theta^*$. A practical arbitrary-start theorem with logarithmic burn-in is a separate extension: it must replace every $Q_l^{(\alpha)}$ in Sections 4.1–4.8

by the burned-in weight $Q_{l,n_0}^{(\alpha)}$ displayed in Section 4.1 and then redo the Poisson, variance-comparison, and misadjustment bookkeeping. The Wasserstein contraction of the augmented chain (Levin et al. 2025, Proposition 1) is the expected tool for this transfer, but the accumulated startup contribution over the averaging window is not part of the theorem proved here.

RR removes the worst remainder, the depth-two transfer recovers the rate. The Poisson boundary remainder is one boundary piece smaller than in the single-step PR case (because $\mathcal{Q}_{n-1}^{\text{RR}} = 0$ removes one of the two boundary terms); the misadjustment is **not** improved by RR at the level of the depth-two analysis used here, because $J^{(2)} + H^{(2)}$ are bounded uniformly in step size at order $\alpha^{3/2}$ and only $J^{(1)}$ enjoys RR-cancellation. The $\log^{3/4}(n) n^{-1/4}$ leading rate is therefore governed by the **martingale** Berry–Esseen, with the misadjustment matching but not strictly subleading. A genuine RR-cancellation at the $J^{(2)} + H^{(2)}$ level would push the misadjustment to $\sqrt{n}\alpha^2 = O(n^{-1/2})$, strictly subleading – the open thread of Section 4.9.

No improvement over single-step PR rate. The Berry–Esseen rate $\log^{3/4}(n) n^{-1/4}$ matches that of Samsonov et al. (2025, Theorem 1) for the single-step PR estimator at the diminishing schedule $\alpha_k \asymp k^{-3/4}$. Richardson–Romberg extrapolation in the **constant**-step regime achieves the same rate without diminishing schedule, at the cost of running two coupled trajectories with step sizes α and 2α .